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Stoeckius

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(54) **ELECTROPHORETIC SYSTEM AND METHOD FOR ANALYTE CAPTURE**

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(57) **ABSTRACT**

An electrophoretic system is provided for analyte capture from a biological sample. The electrophoretic system can be used to permeabilize the sample to allow analytes to be released from the sample. For example, the sample can be contacted with capture probes attached to a substrate, and an electric field created by the electrophoretic system can cause analytes to be released from the cell, and effectively migrate toward and bind to the capture probes attached to the substrate.

20 Claims, 29 Drawing Sheets

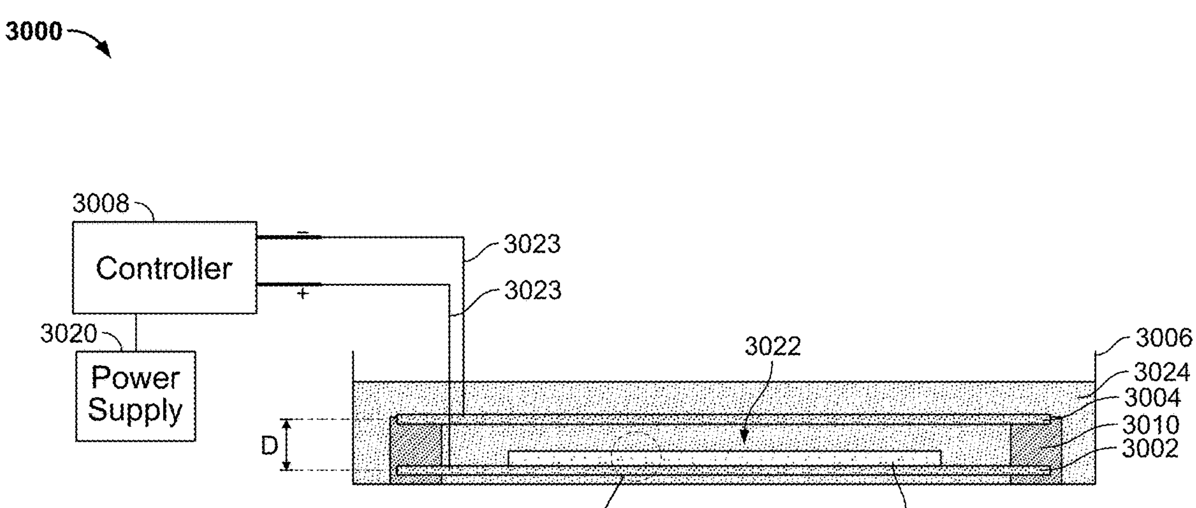

FIG. 3A

(58) **Field of Classification Search**
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USPC 435/971, 204, 205, 40, 287.9, 2, 287.1, 435/287.2, 288.3, 288.4, 7.1; 436/7, 775, 436/151, 205, 501, 516; 204/450, 451, 204/452
See application file for complete search history.

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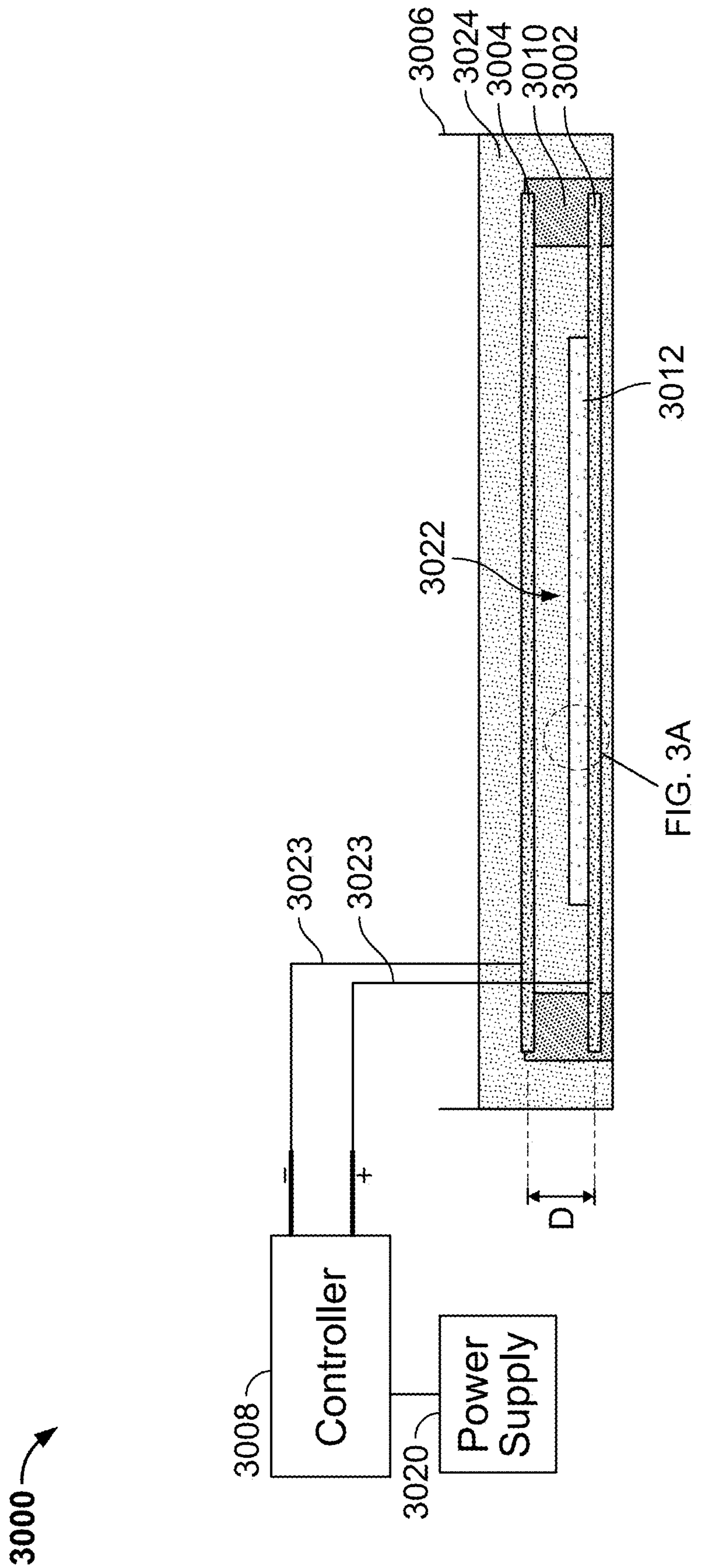

3006
3024
3004
3010
3002
3012
3022
FIG. 3A
3023
3023
−
+
D
Controller
Power Supply
3008
3020
3000

FIG. 1

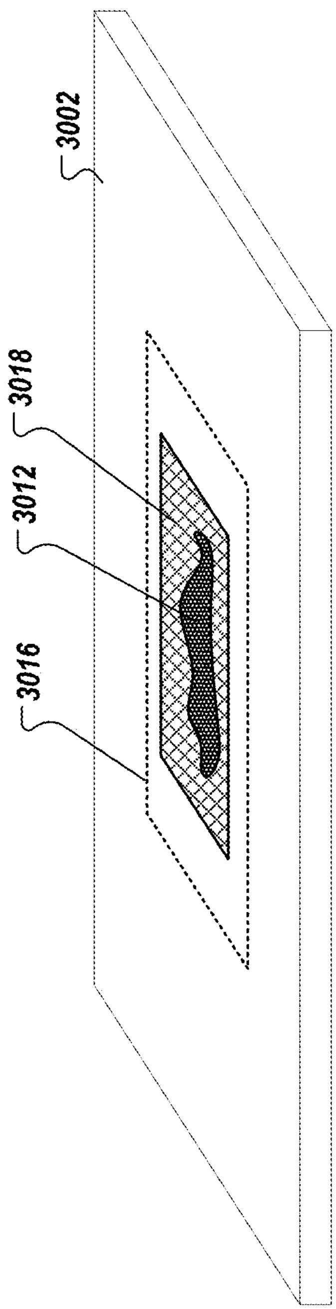


FIG. 2

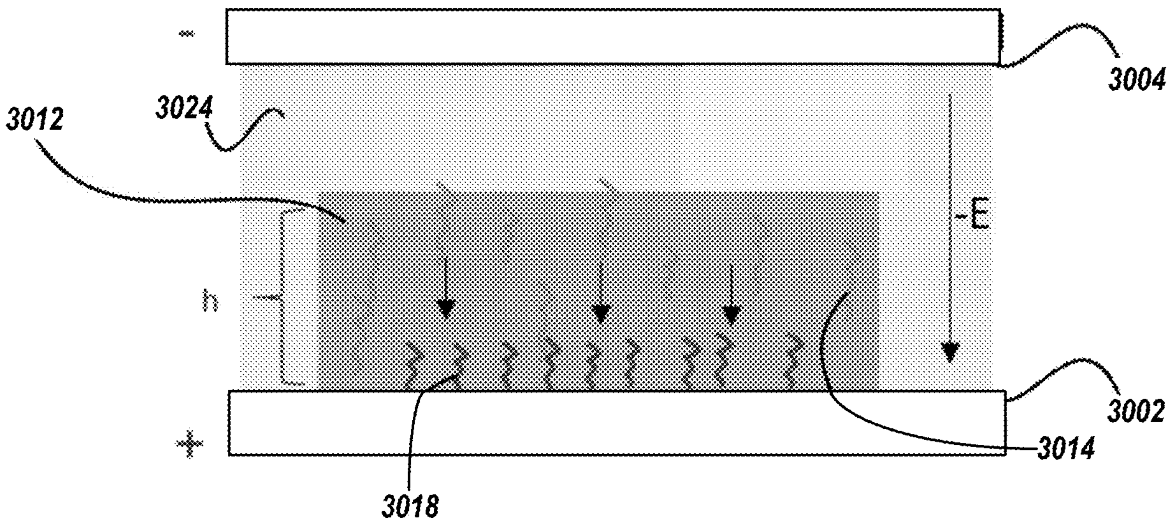

-
3004
3012
3024
-E
h
3002
+
3014
3018

FIG. 3A

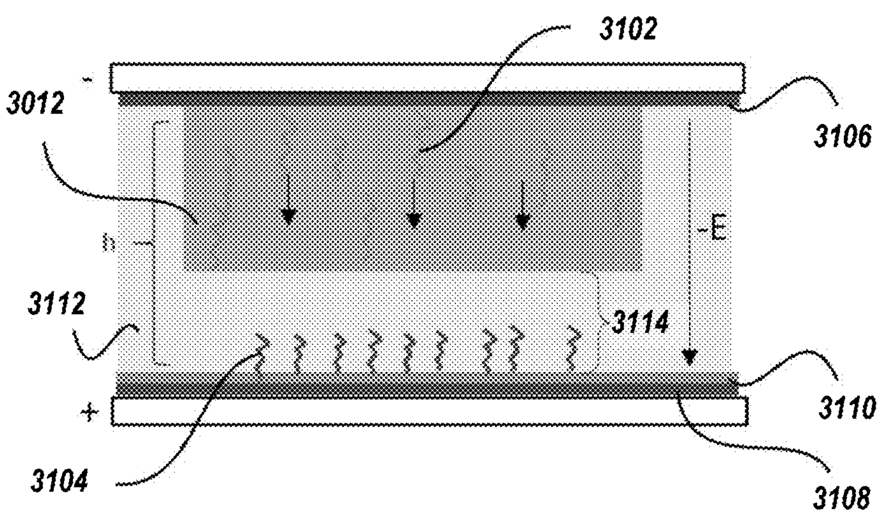
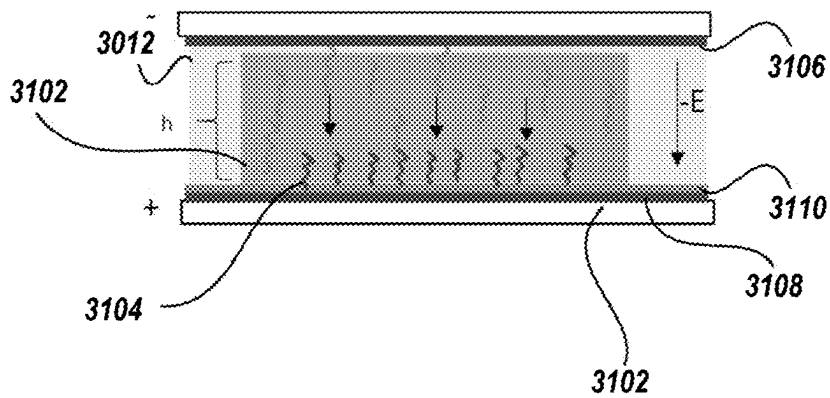
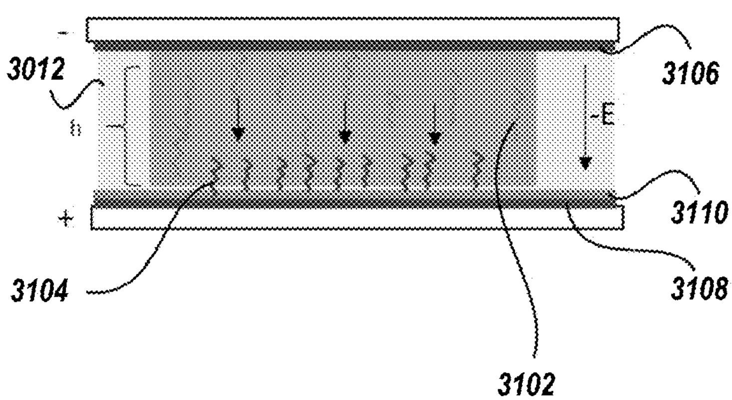

3102
-
3106
3012
-E
h
3112
3114
+
3110
3104
3108

FIG. 3B



3012
3102
h
-E
3106
3110
3108
3104
3102

FIG. 3C



3012
h
-E
3106
3110
3108
3104
3102

FIG. 3D

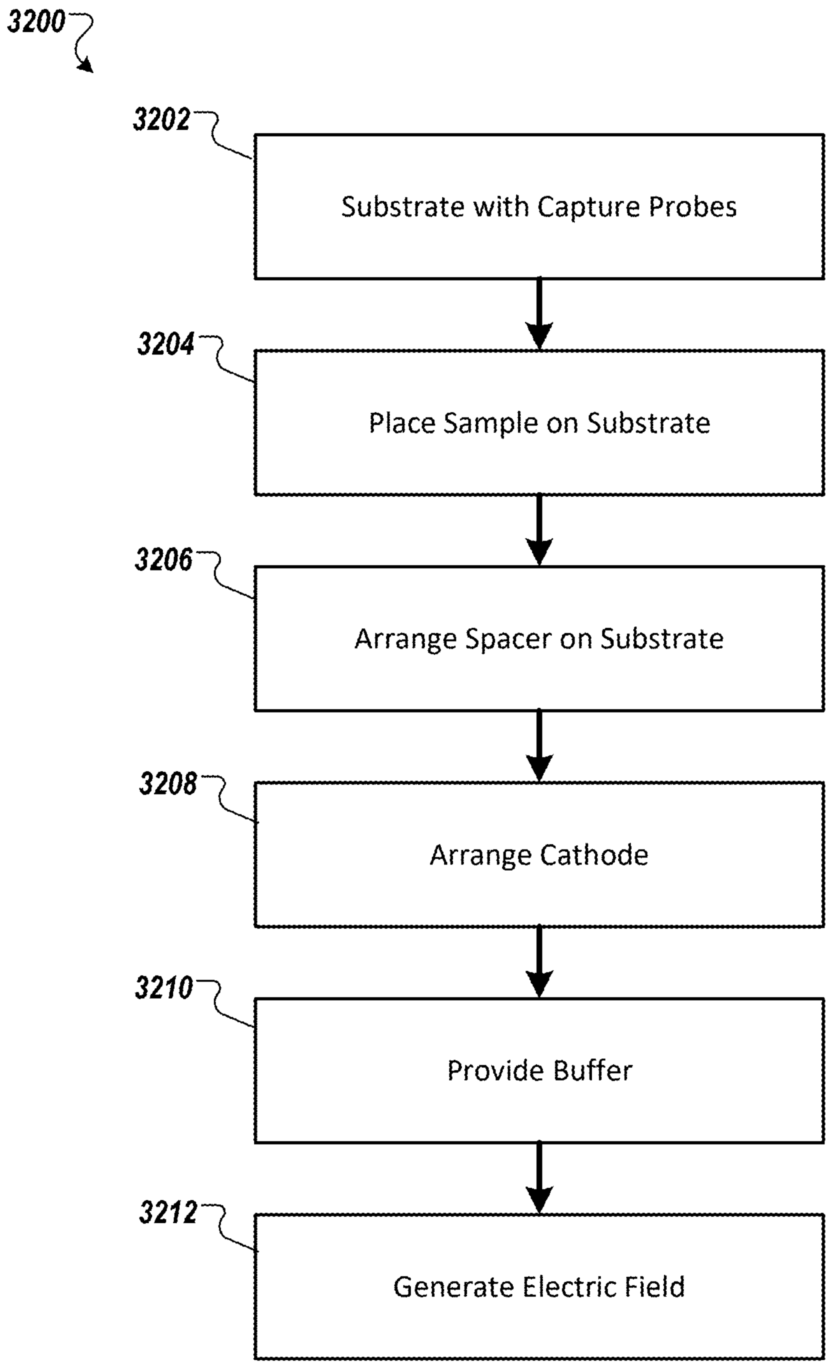

3200
3202
Substrate with Capture Probes
3204
Place Sample on Substrate
3206
Arrange Spacer on Substrate
3208
Arrange Cathode
3210
Provide Buffer
3212
Generate Electric Field

FIG. 4

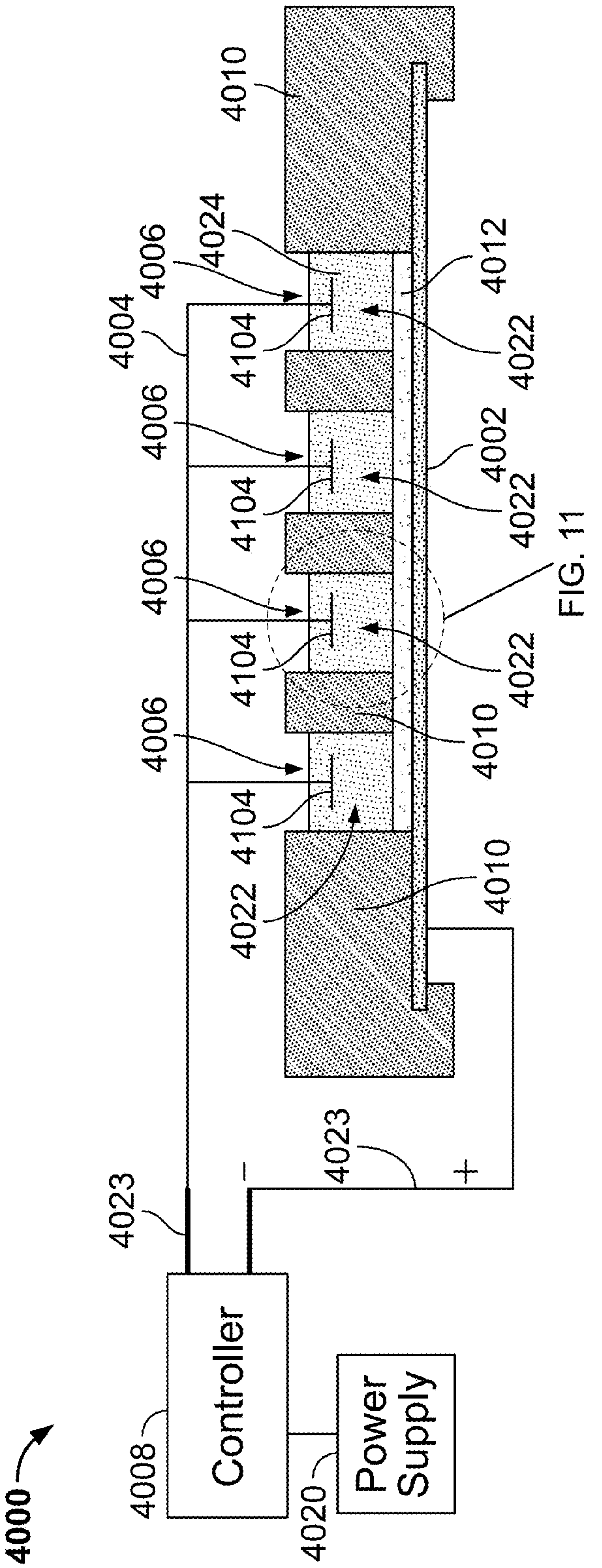
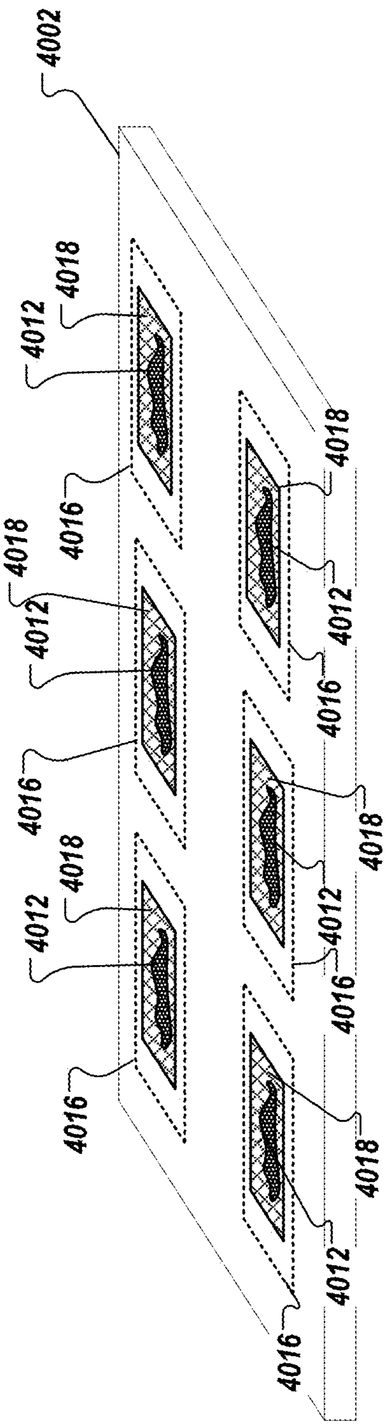

4000
4008
Controller
4020
Power Supply
4023
4023
-
+
4004
4006
4006
4006
4006
4024
4104
4104
4104
4104
4022
4022
4022
4022
4010
4010
4010
4012
4002
FIG. 11

FIG. 5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4012
4018
4016
4012
4018
4016
4012
4018
4016
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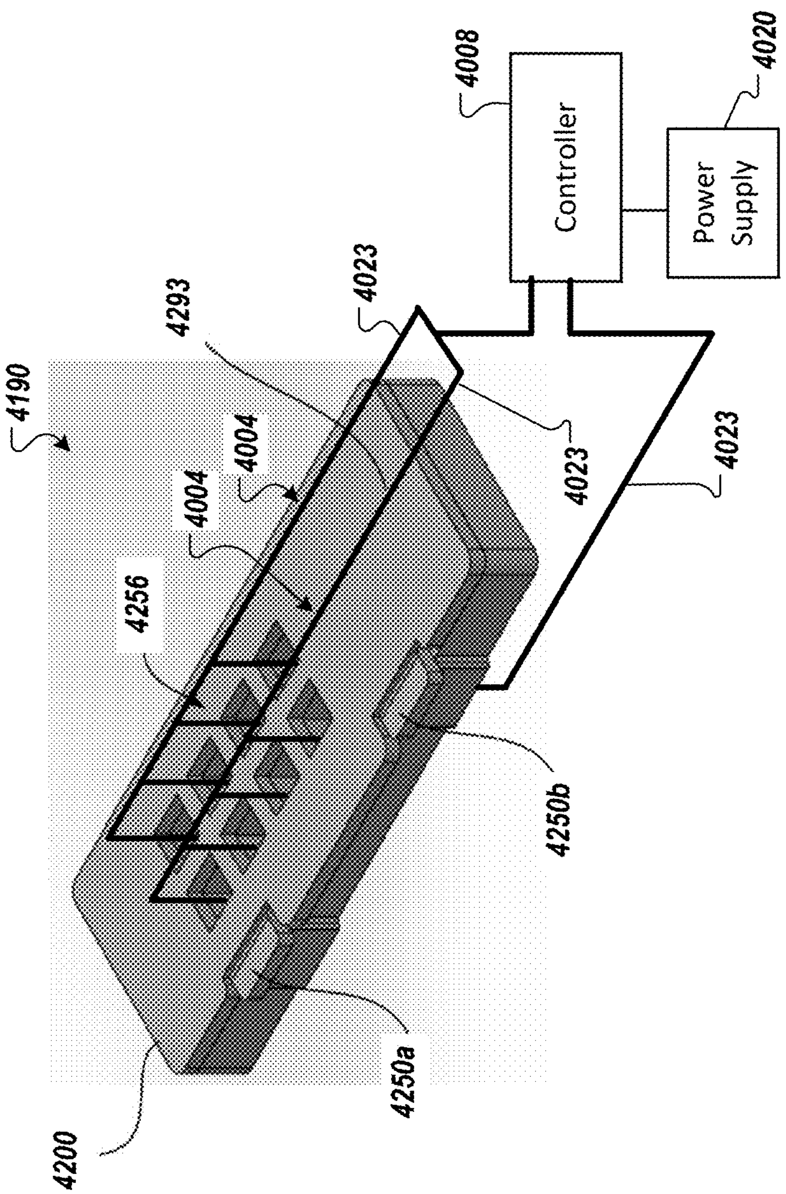
4012
4016
4018
4012
4016
4018
4012
4016

FIG. 6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4004
4256
4293
4023
4023
4023
4008
Controller
4020
Power Supply
4250b
4250a
4200

FIG. 7

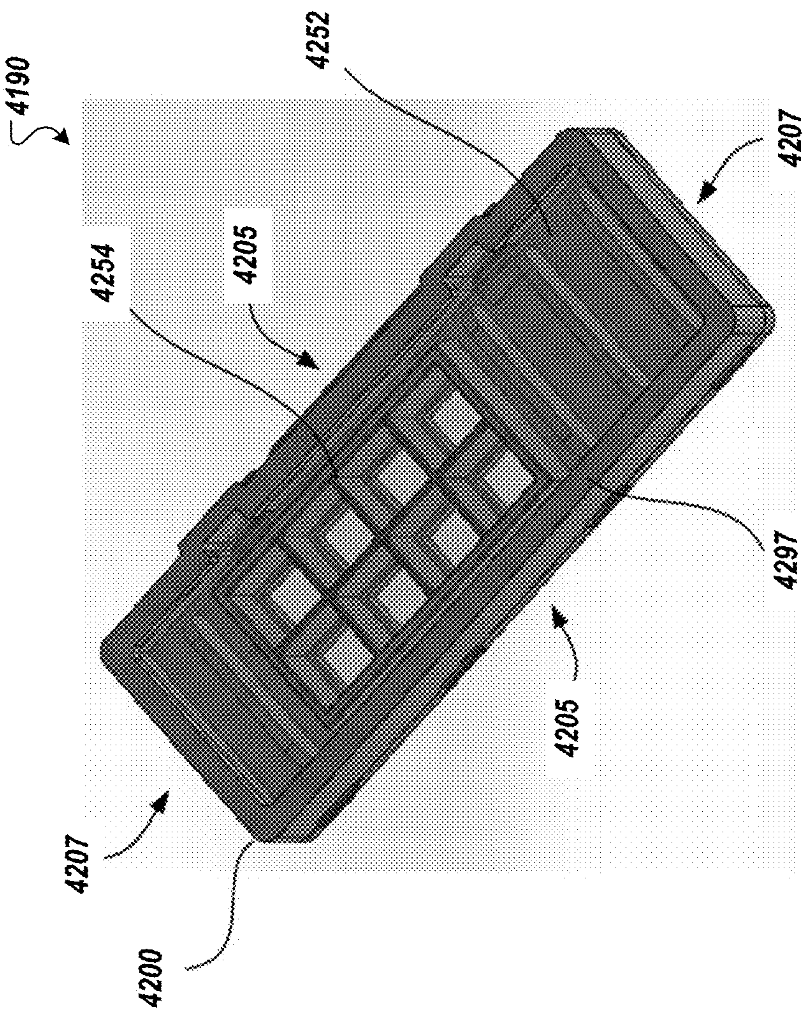

FIG. 8

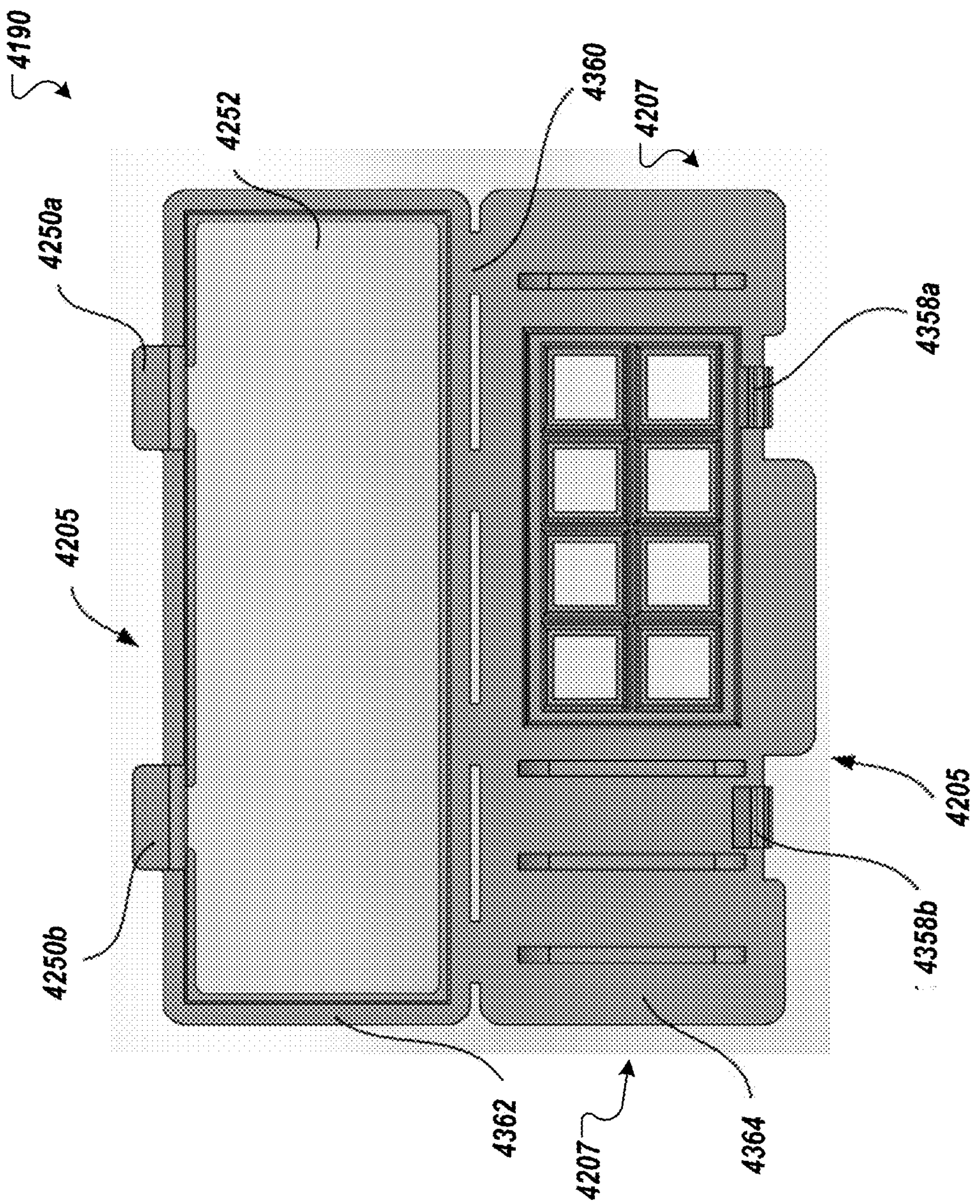
4190
4252
4360
4207
4250a
4358a
4205
4205
4250b
4358b
4362
4207
4364

FIG. 9A

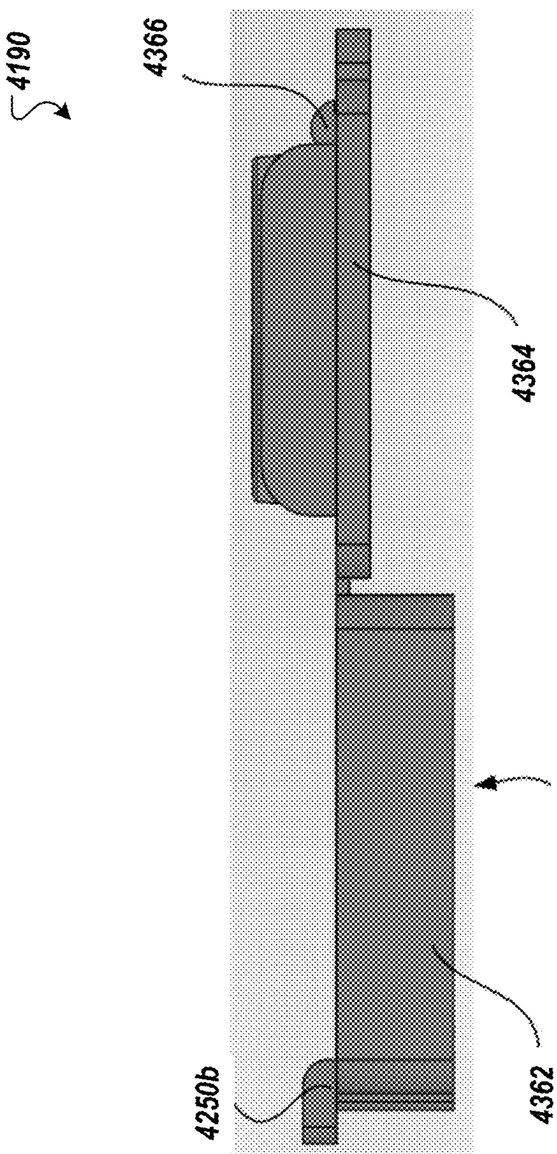

FIG. 9B

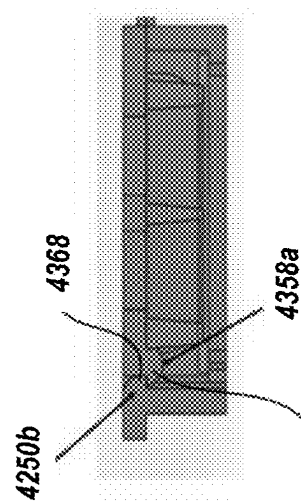

FIG. 9C

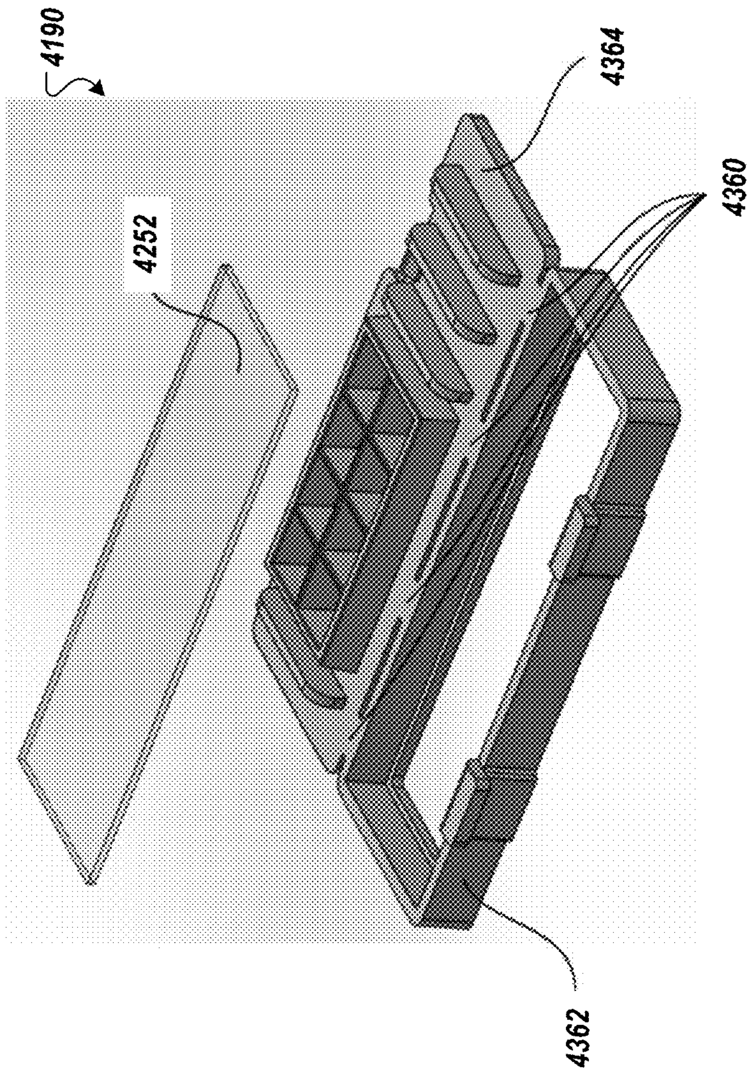

FIG. 10A

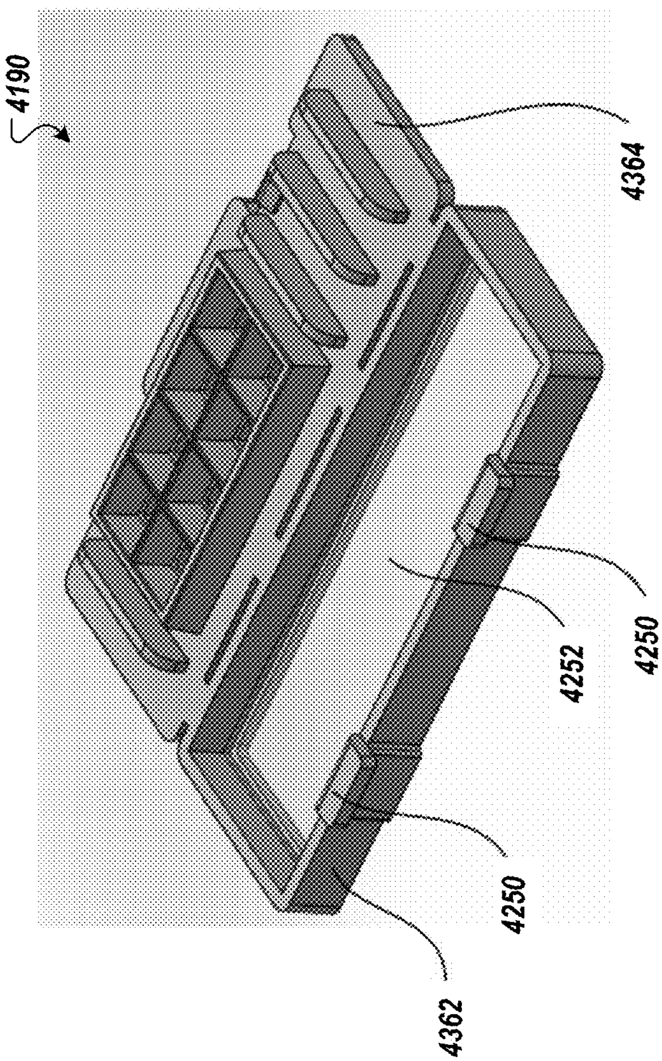

FIG. 10B

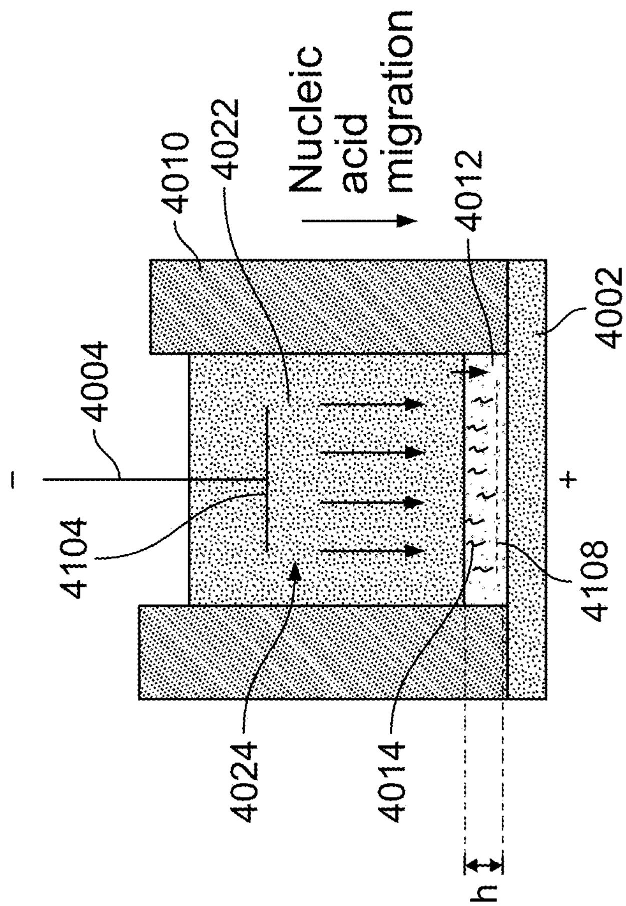
4010
4022
Nucleic
acid
migration
4012
4004
–
+
4002
4104
4108
4024
4014
h

FIG. 11

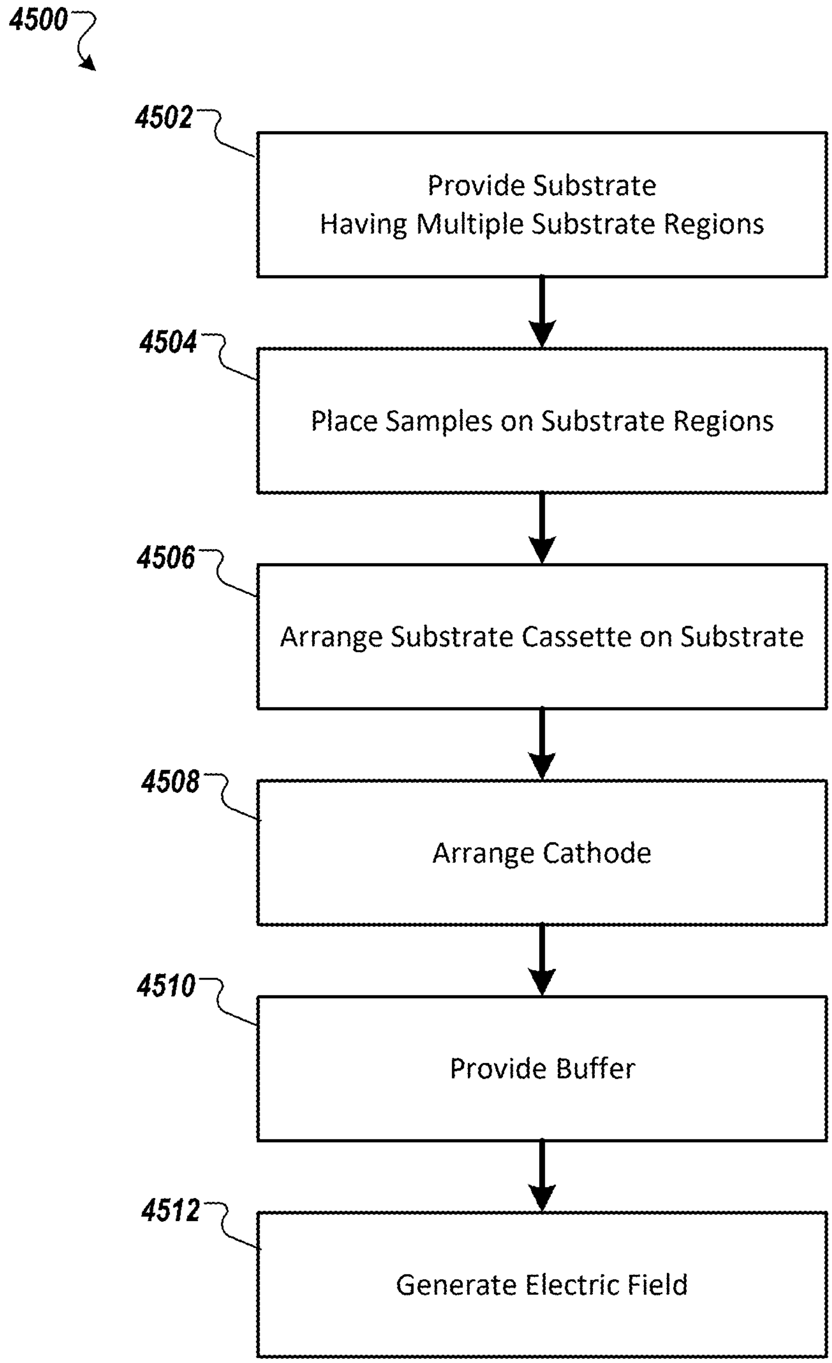
4500
4502
Provide Substrate
Having Multiple Substrate Regions
4504
Place Samples on Substrate Regions
4506
Arrange Substrate Cassette on Substrate
4508
Arrange Cathode
4510
Provide Buffer
4512
Generate Electric Field

FIG. 12

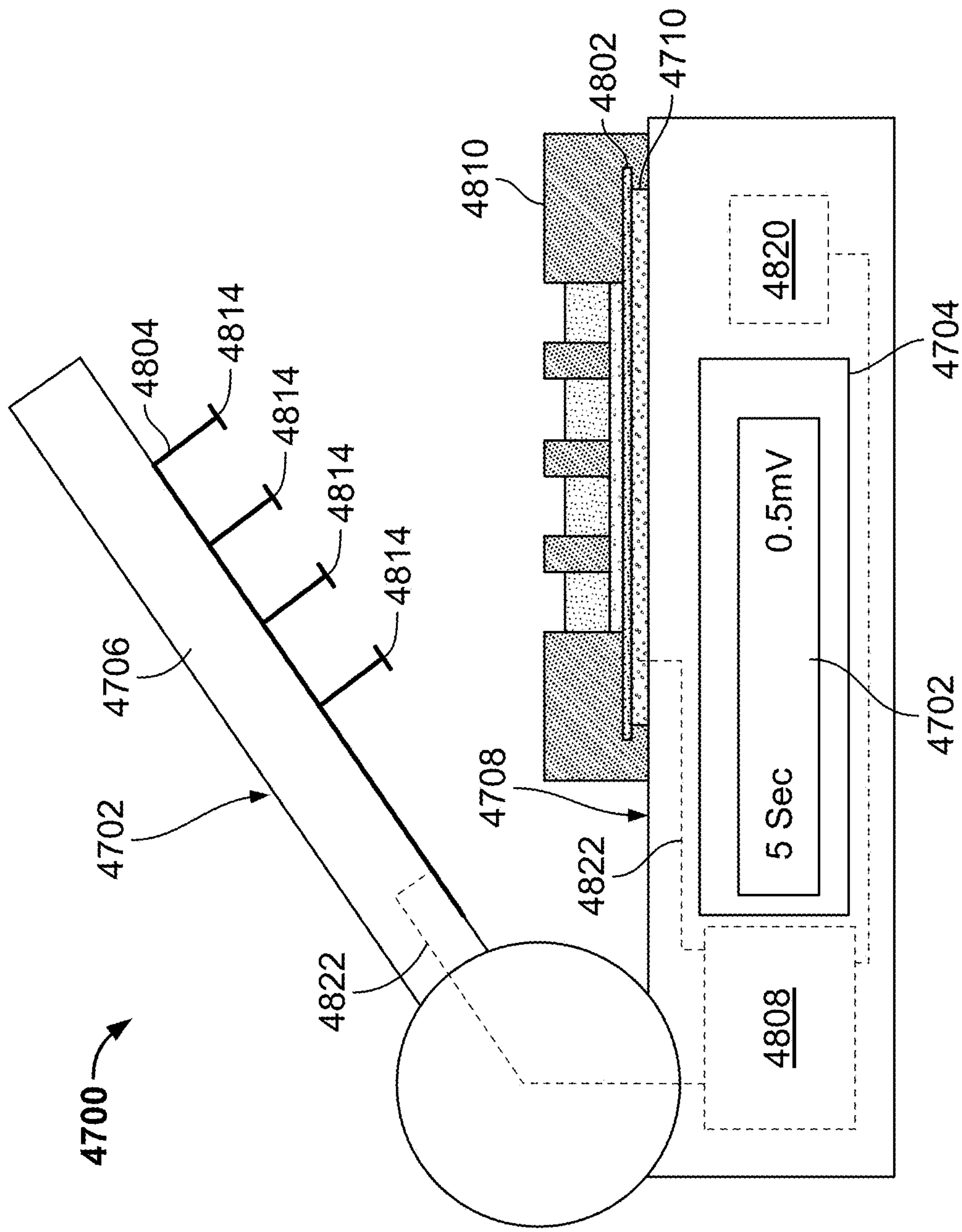
4700
4702
4706
4804
4814
4814
4814
4814
4822
4810
4802
4710
4708
4822
4820
4704
0.5mV
5 Sec
4702
4808

FIG. 13

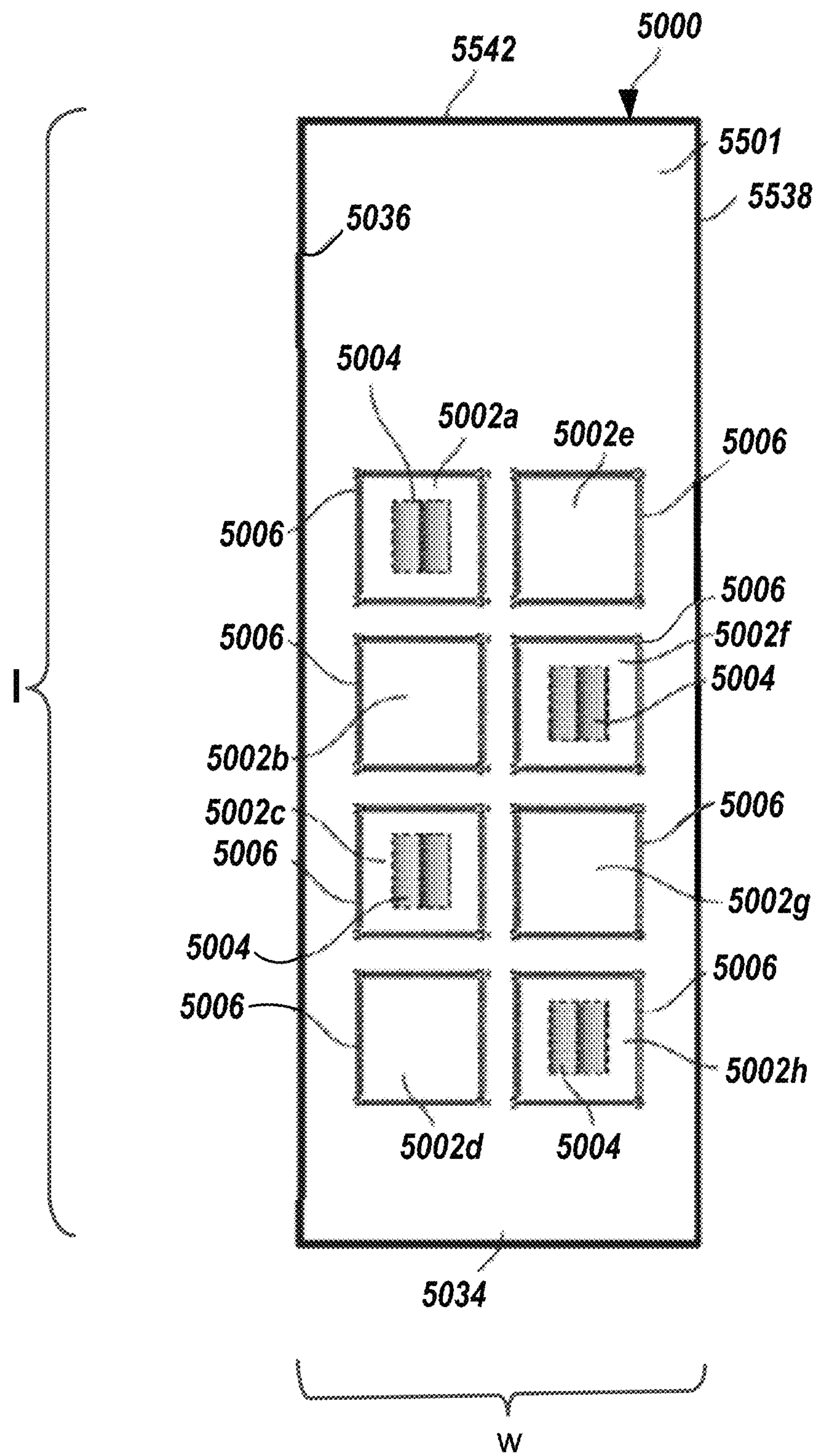
5000
5542
5501
5538
5036
5004
5002a
5002e
5006
5006
5006
5006
5002f
5004
5002b
5002c
5006
5006
5002g
5004
5006
5006
5002h
5002d
5004
5034
l
w

FIG. 14

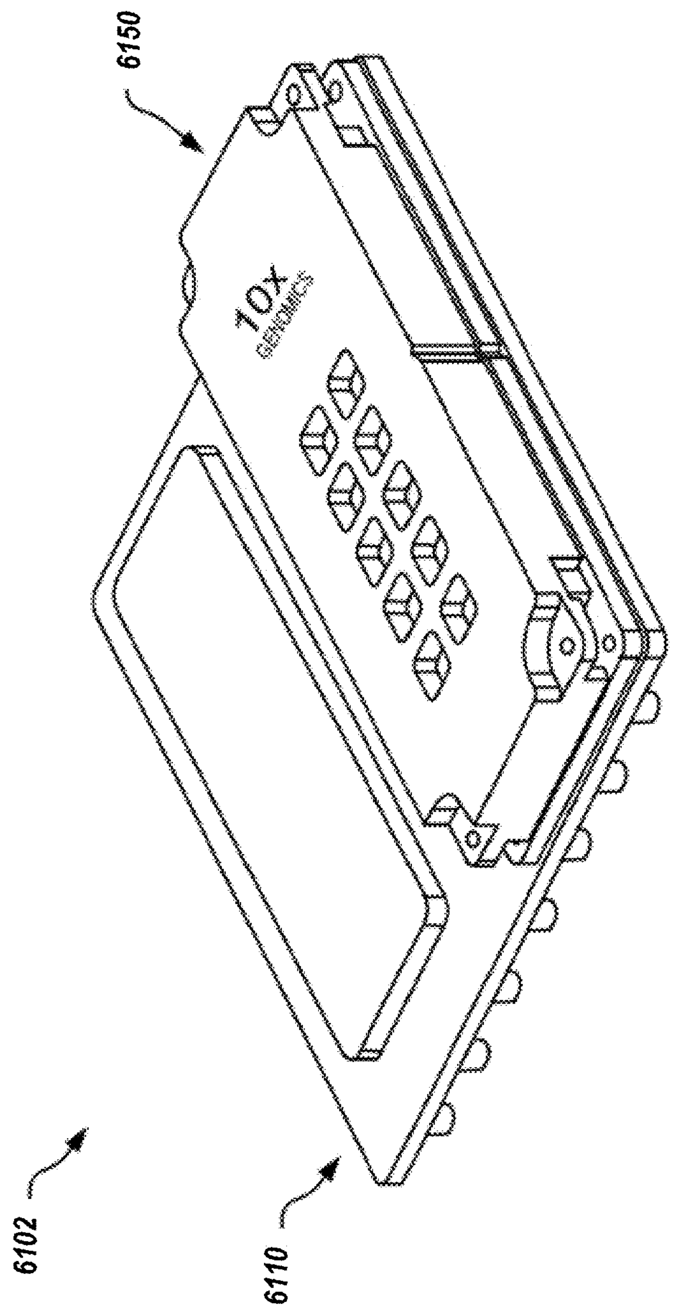

FIG. 15A

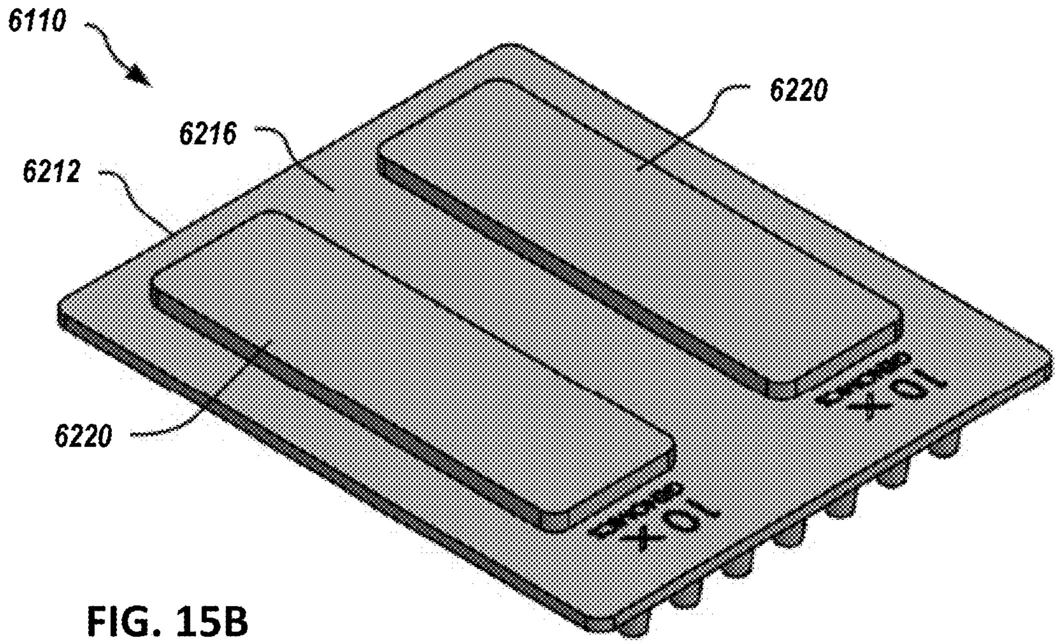

FIG. 15B

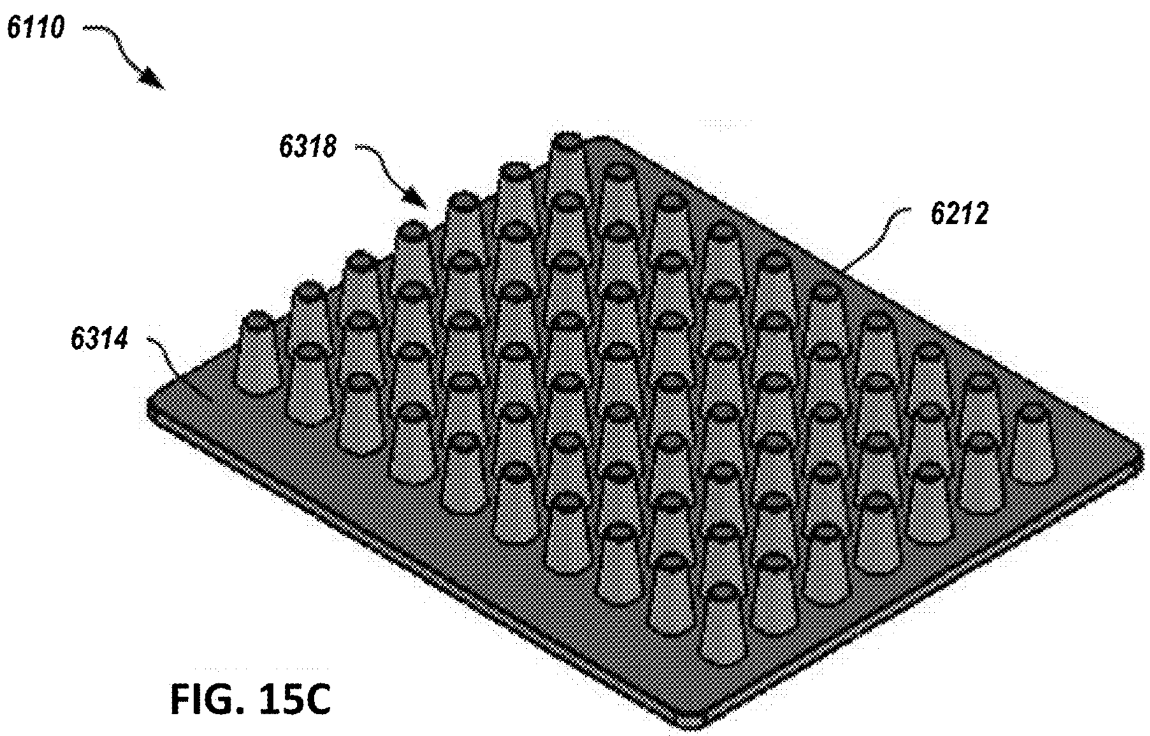

FIG. 15C

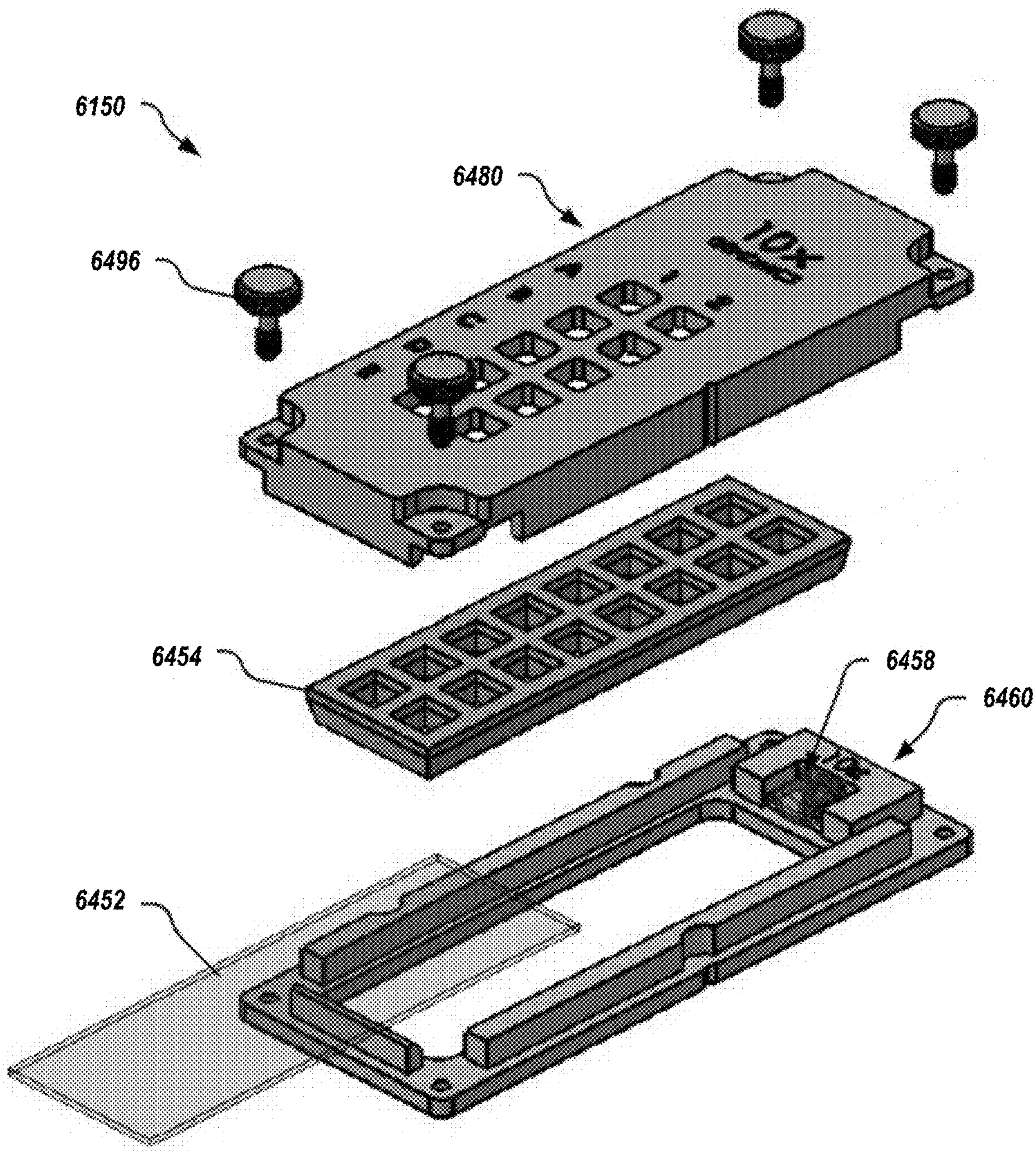

FIG. 16

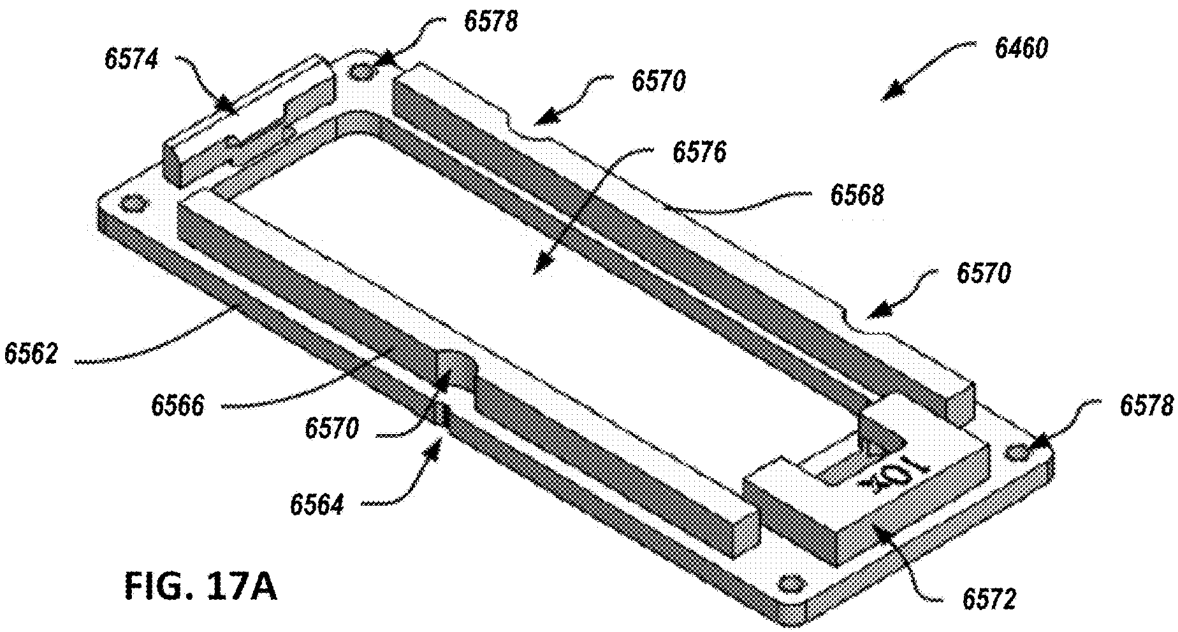

FIG. 17A

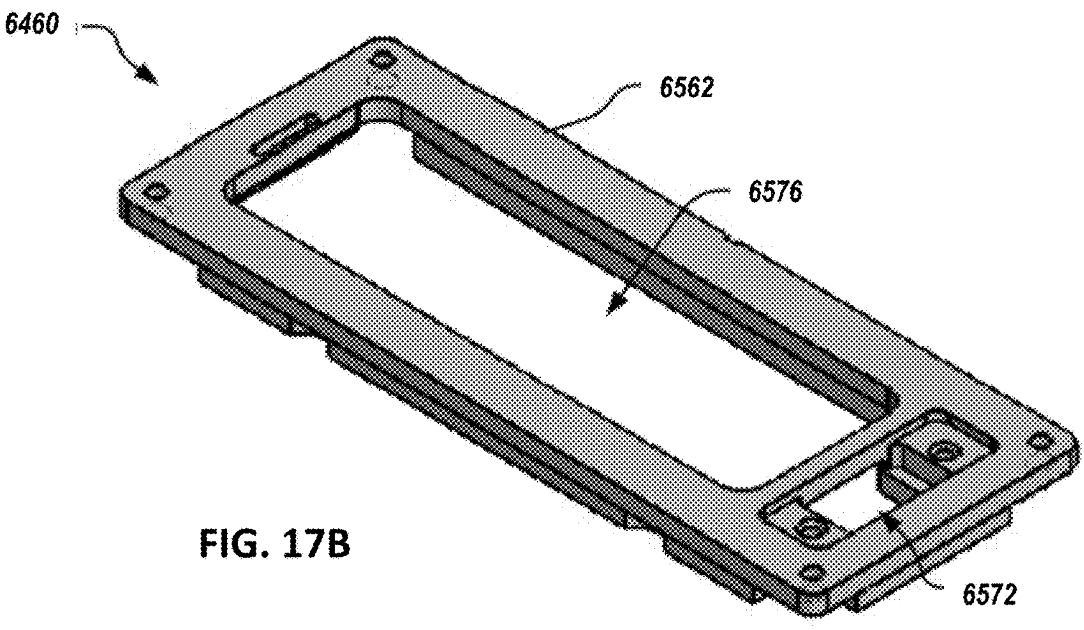

FIG. 17B

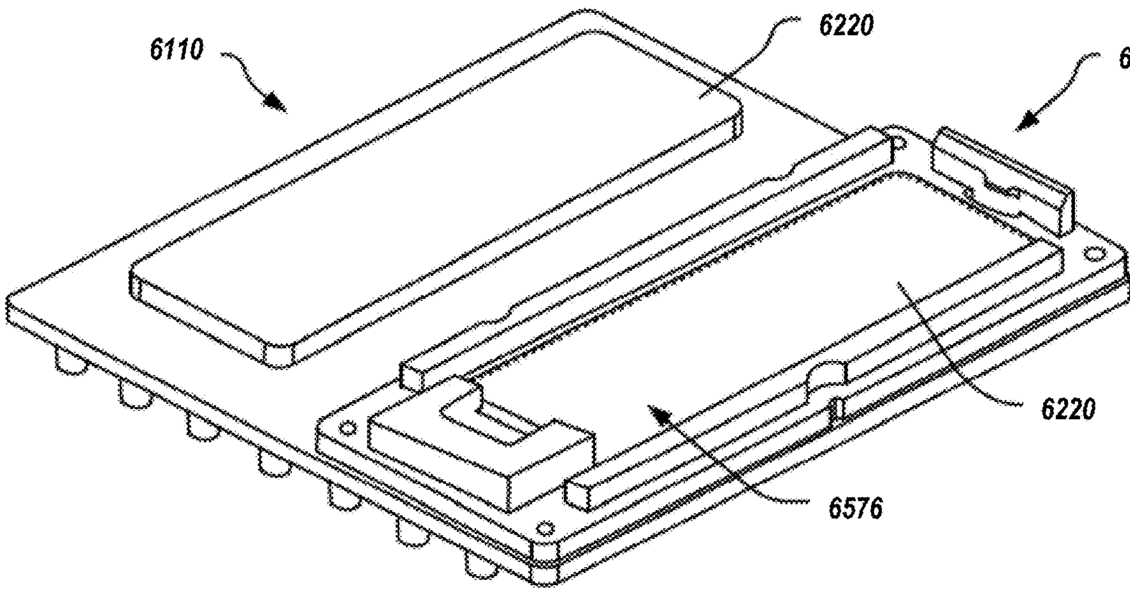

FIG. 18

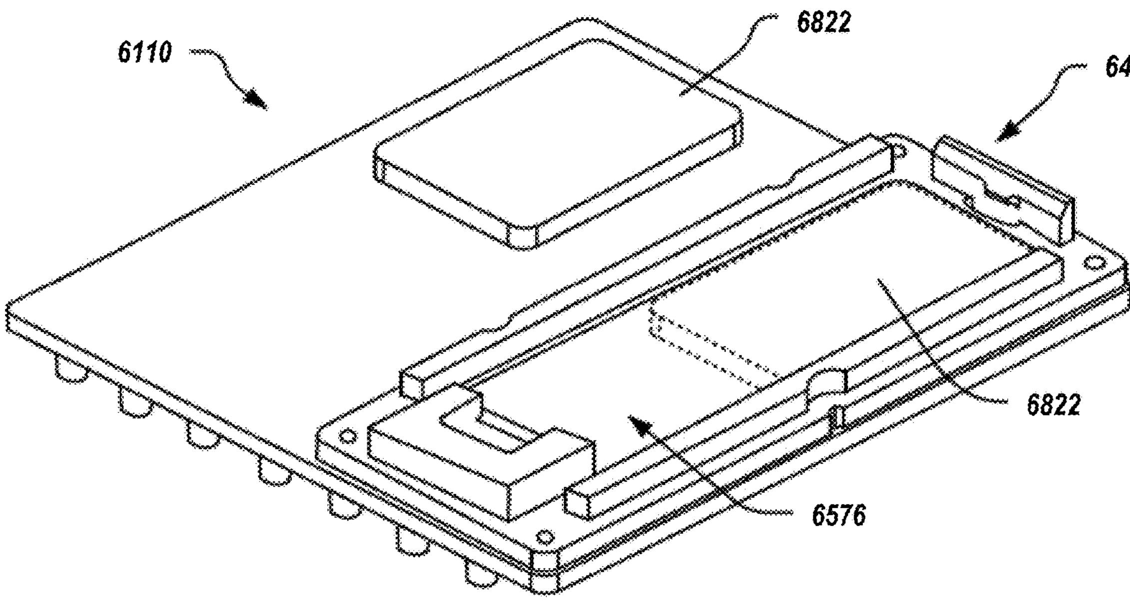

FIG. 19

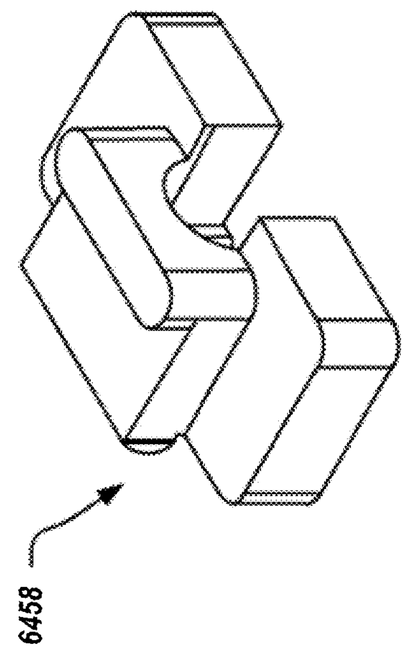

FIG. 20B

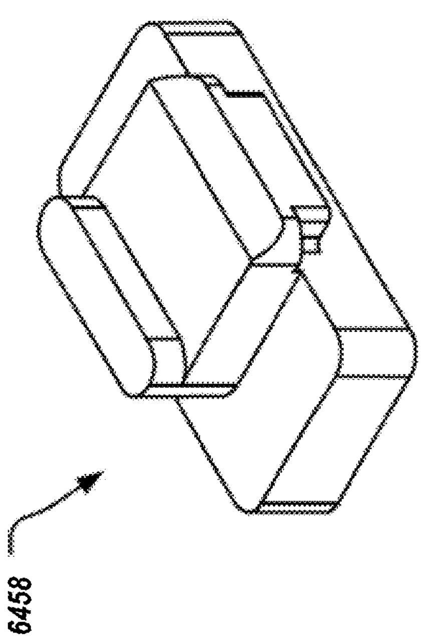

FIG. 20A

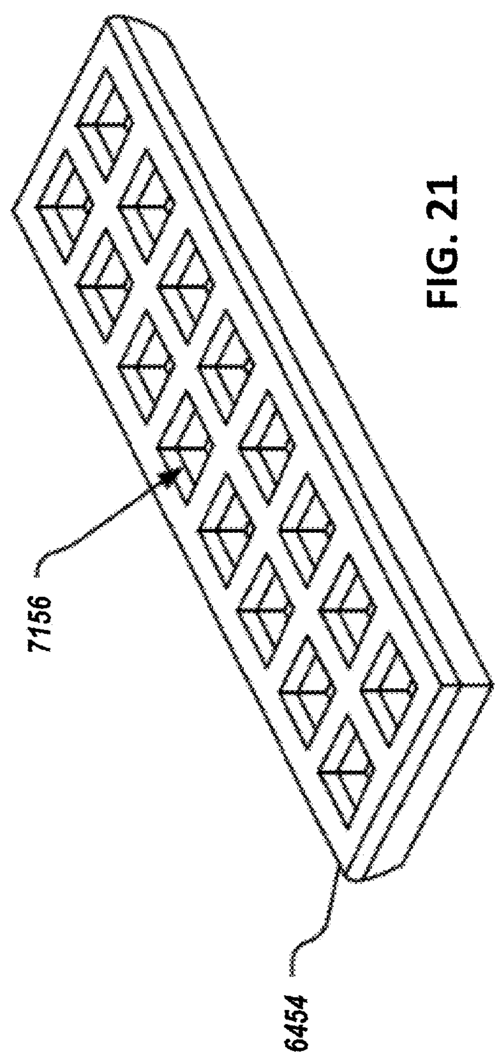

FIG. 21

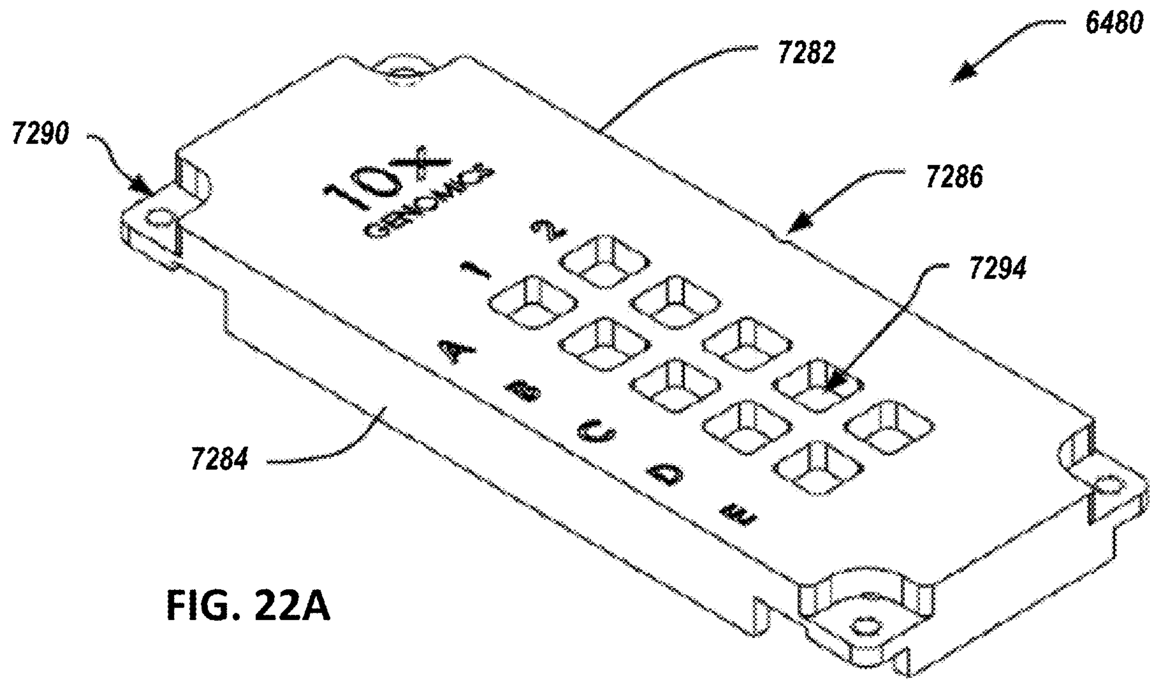

FIG. 22A

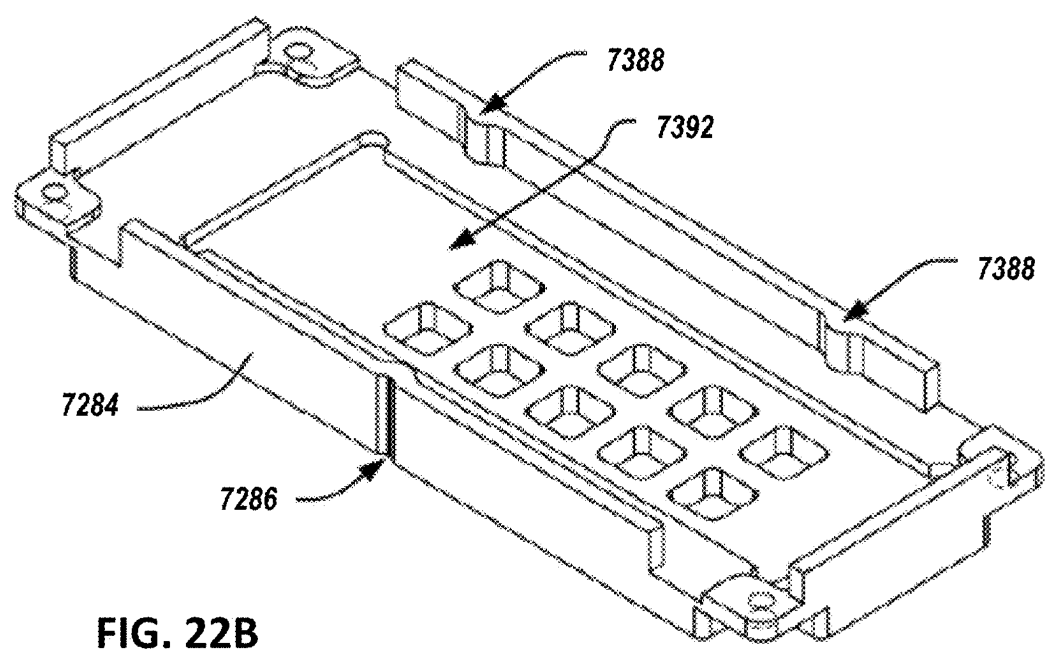

FIG. 22B

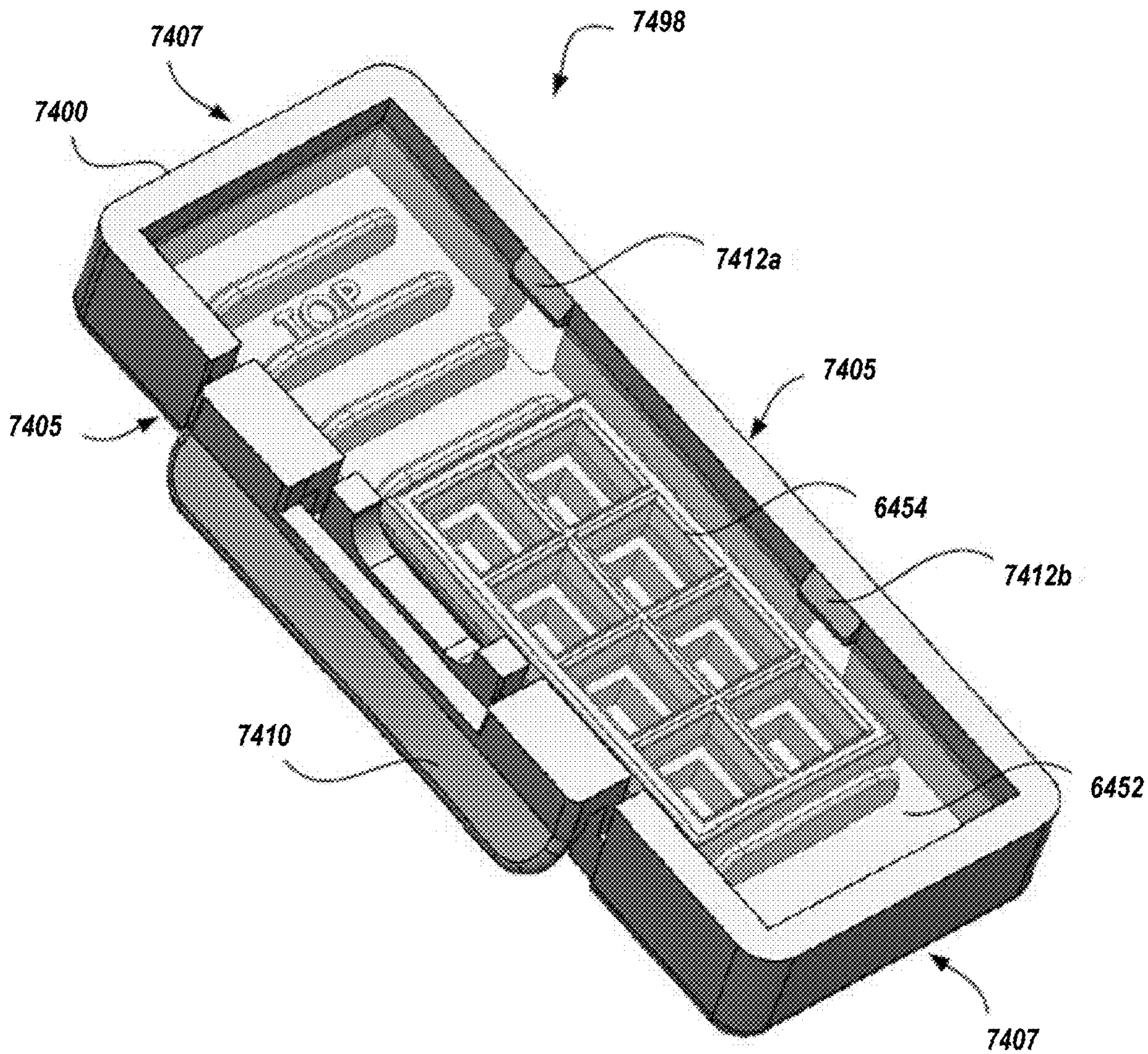

FIG. 23A

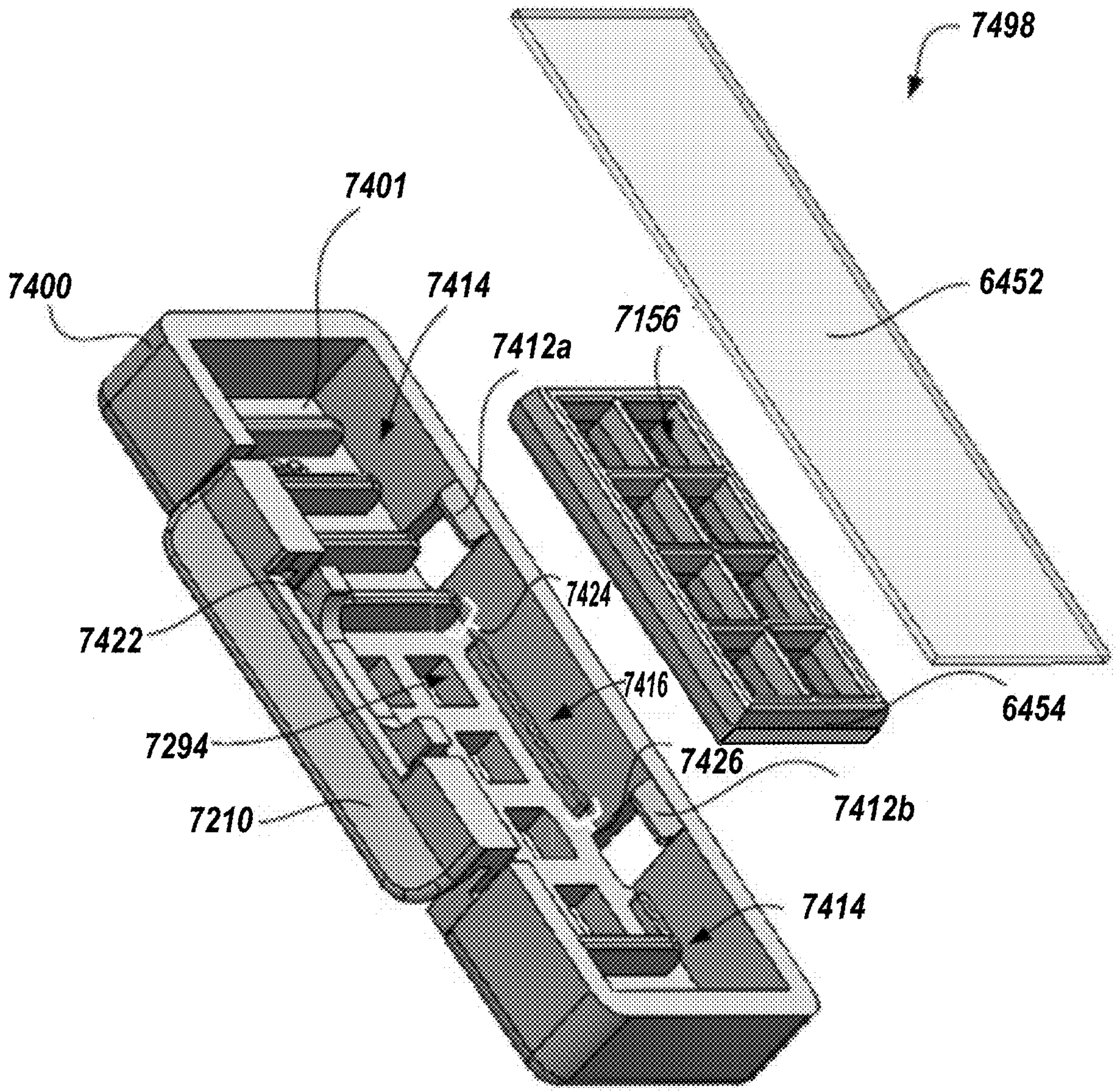
7498
7401
7400
7414
6452
7156
7412a
7424
7422
7416
6454
7294
7426
7210
7412b
7414

FIG. 23B

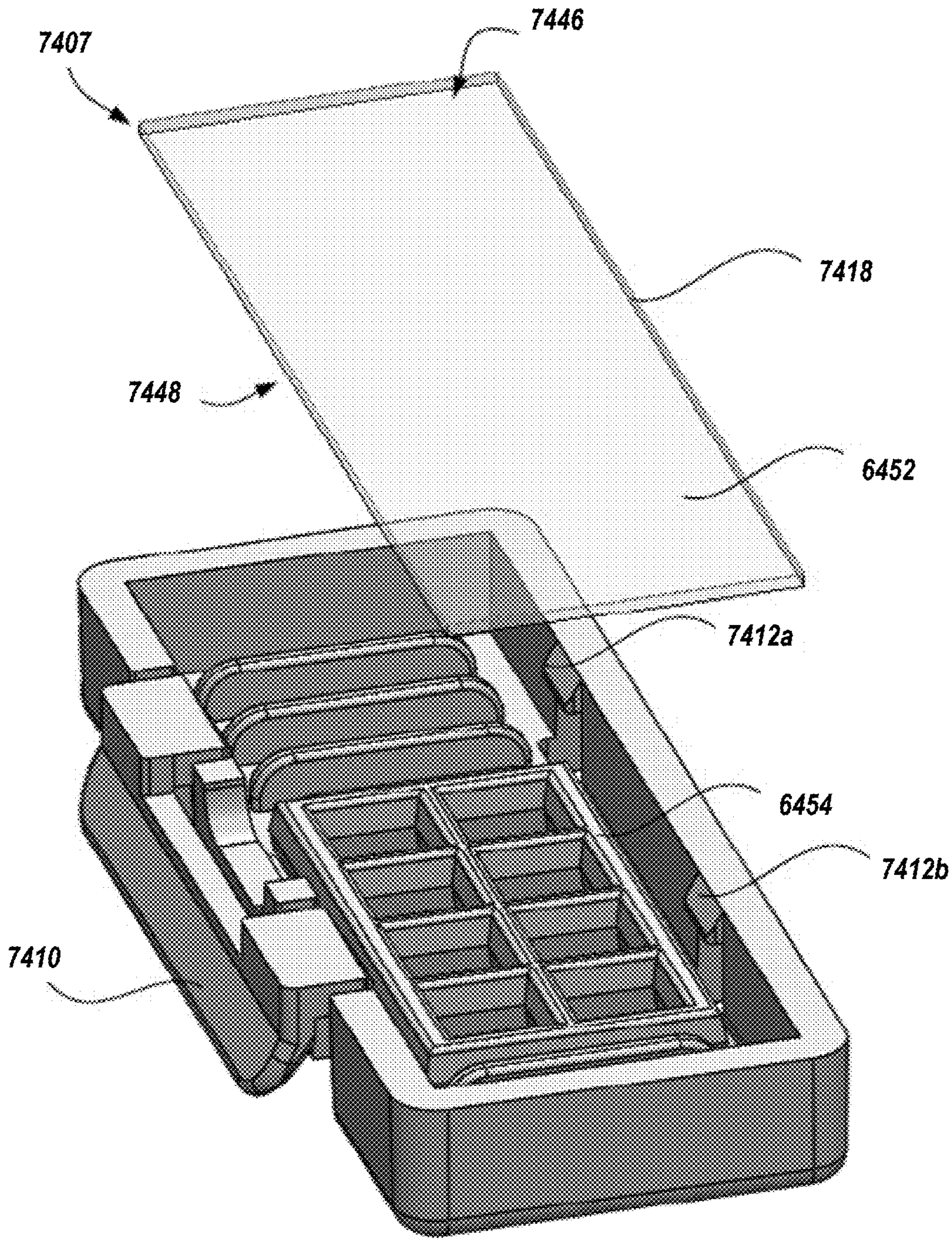

FIG. 23C

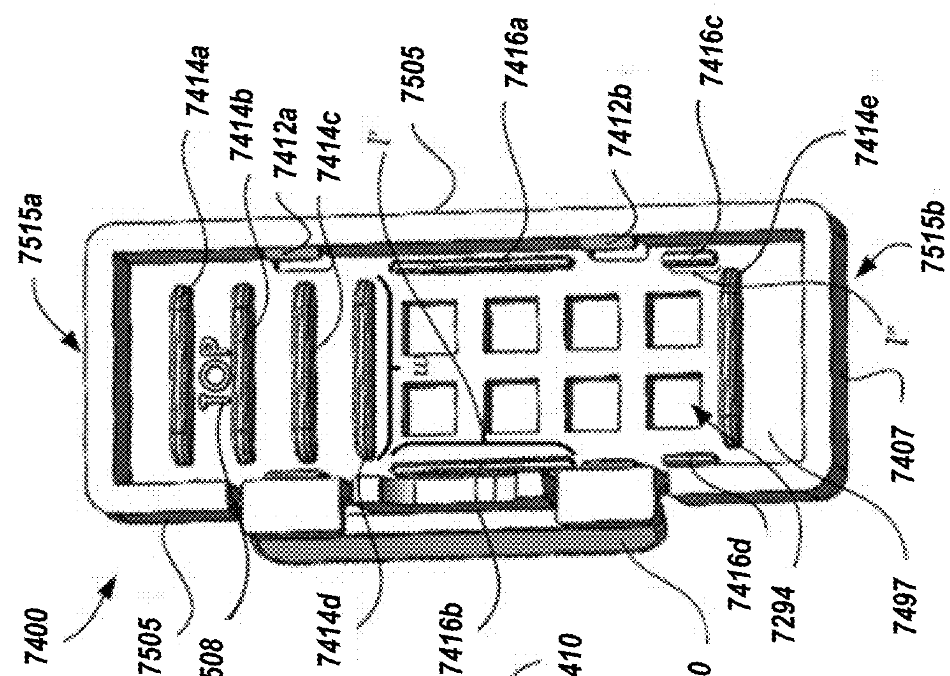

FIG. 24B

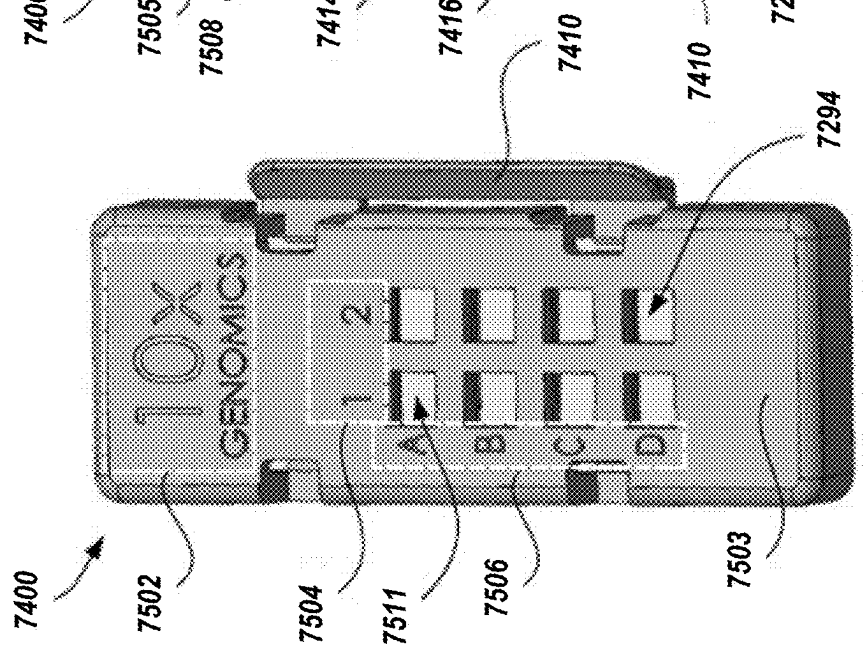

FIG. 24A

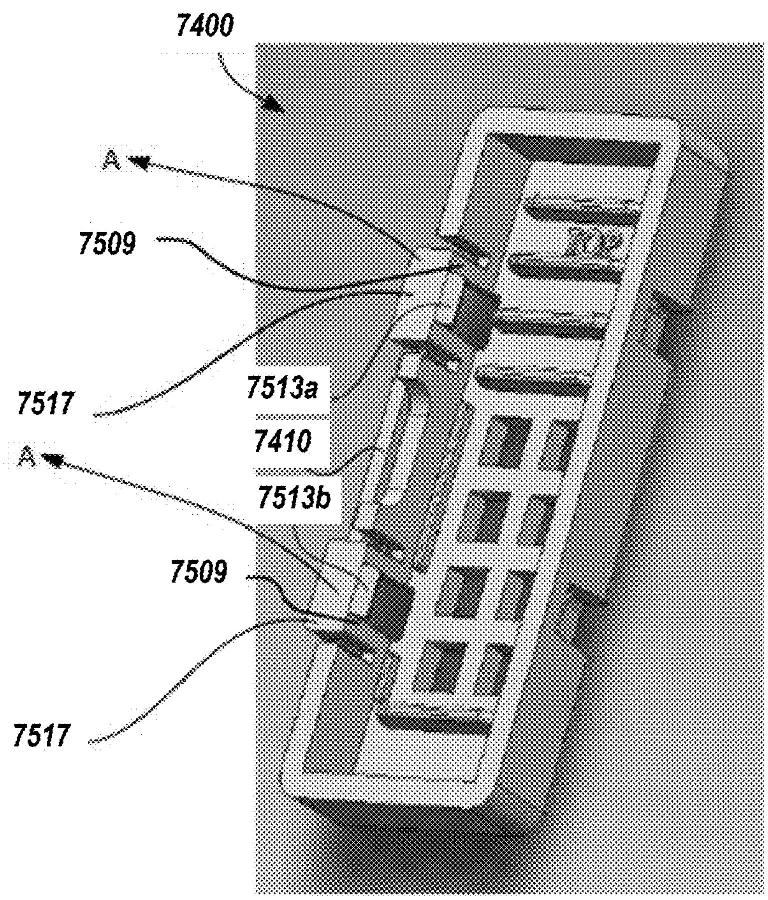

FIG. 24C

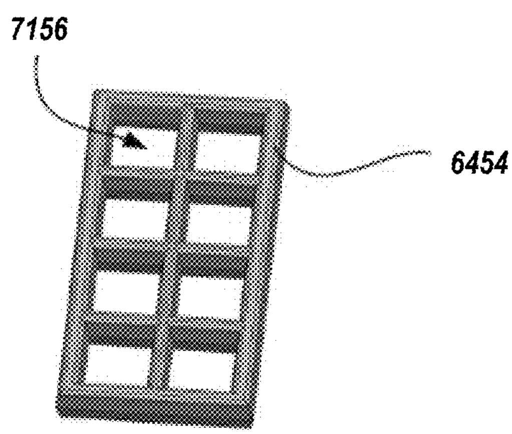

FIG. 25

ELECTROPHORETIC SYSTEM AND METHOD FOR ANALYTE CAPTURE

CROSS-REFERENCE TO RELATED APPLICATION

This application claims the benefit of U.S. Patent Application Ser. No. 62/962,559, titled ELECTROPHORETIC SYSTEM AND METHOD FOR PREPARING SAMPLE, filed Jan. 17, 2020, the disclosure of which is hereby incorporated by reference in its entirety.

BACKGROUND

Cells within a tissue have differences in cell morphology and/or function due to varied analyte levels (e.g., gene and/or protein expression) within the different cells. The specific position of a cell within a tissue (e.g., the cell's position relative to neighboring cells or the cell's position relative to the tissue microenvironment) can affect, e.g., the cell's morphology, differentiation, fate, viability, proliferation, behavior, signaling, and cross-talk with other cells in the tissue.

Spatial heterogeneity has been previously studied using techniques that typically provide data for a handful of analytes in the context of intact tissue or a portion of a tissue (e.g., tissue section), or provide significant analyte data from individual, single cells, but fails to provide information regarding the position of the single cells from the originating biological sample (e.g., tissue).

Various methods have been used to capture analytes from a biological sample for analyzing analyte data in the sample. In some applications, a biological sample can be permeabilized to facilitate transfer of analytes out of the sample, and/or to facilitate transfer of species (such as capture probes) into the sample. If a sample is not permeabilized sufficiently, the amount of analyte captured from the sample may be too low to enable adequate analysis.

SUMMARY

This document generally relates to electrophoretic apparatuses, systems, and methods for capturing analytes from a sample, such as a biological sample.

Some embodiments described herein include an electrophoretic system for analyte capture from a biological sample, such as a cell or a tissue sample including a cell. The electrophoretic system can be used to permeabilize the sample to allow analytes to be released from the sample (e.g., the cell therein). For example, the sample can be contacted with capture probes attached to a substrate (e.g., a surface of the substrate), and an electric field created by the electrophoretic system can cause analytes to be released from the cell, and effectively migrate toward and bind to the capture probes attached to the substrate. Loss of spatial resolution can occur when analytes migrate from a sample to capture probes (e.g., feature array) and a component of diffusive migration occurs in a transverse (e.g., lateral) direction, approximately parallel to the surface of the substrate on which the sample is mounted. The electrophoretic system described herein can actively direct analytes released from a cell to the capture probes, thereby improving spatial resolution by eliminating or reducing such diffusive migration in the transverse direction.

In some implementations, a substrate (e.g., slide) on which a sample (e.g., cell, tissue, etc.) is placed can include a conductive material and used as an anode in the electrophoretic system described herein. An electrode can be placed to be spaced apart from the substrate, and used as a cathode. A non-conductive spacer can be arranged between the substrate and the electrode. The substrate, the electrode, and the spacer can be at least partially immersed in a buffer. The electrophoretic system can apply a voltage between the electrodes (i.e., the substrate comprising the anode, and the electrode as the cathode) to cause analytes to release from the sample and migrate to capture probes attached to the substrate.

In some implementations, the electrophoretic system described herein can include a variety of substrate cassettes configured for analyzing multiple samples. For example, a substrate cassette can be configured to engage a substrate that has multiple substrate regions. The substrate cassette can include a plurality of apertures corresponding to the substrate regions of the substrate, respectively. The plurality of apertures can be used as a plurality of chambers for receiving buffers, respectively. The substrate can be made of a conductive material and used as a common anode for the plurality of chambers. A cathode can be configured to include a plurality of electrode plates or pins that can extend into the chambers at least partially filled with buffers, respectively. The electrophoretic system can apply a voltage between the substrate (as the anode) and the cathode (including the electrode plates or pins) to cause analytes to release from the samples in each of the substrate regions, and migrate to a capture probe provided at each of the substrate regions of the substrate.

Particular embodiments described herein include an electrophoretic system for migrating analytes in a biological sample. The system includes a substrate, a cathode, a buffer chamber, and a controller. The substrate may include a substrate region configured to place a capture probe thereon. The substrate region may be configured to receive the biological sample containing analytes. The substrate may be configured to be usable as an anode. The cathode may be spaced apart from the substrate. The buffer chamber may be disposed between the substrate and the cathode and configured to contain a buffer. The controller may be configured to generate an electric field between the substrate and the cathode such that the analytes in the biological sample migrate toward the capture probe on the substrate.

In some implementations, the system can optionally include one or more of the following features. The substrate may include a conductive material. The substrate may be coated with a conductive material. The conductive material may include at least one of tin oxide (TO), indium tin oxide (ITO), a transparent conductive oxide (TCO), aluminum doped zinc oxide (AZO), or fluorine doped tin oxide (FTO). The substrate may include an array of substrate regions configured to place capture probes thereon. The capture probe may be immobilized on the substrate region. The system may include a spacer disposed between the substrate and the cathode to define the buffer chamber. The buffer may include a permeabilization reagent. The system may include a power supply, and electrical wires connecting the power supply to the substrate and the cathode. The system may include a substrate cassette configured to hold the substrate and include a plurality of apertures configured to define a plurality of buffer chambers on the substrate.

Particular embodiments described herein include a method for migrating analytes in a biological sample to a substrate. The method may include placing the biological sample in contact with a capture probe on a substrate, the biological sample including analytes; arranging a cathode relative to the substrate at a distance; providing a buffer between the cathode and the biological sample on the substrate; and generating an electric field between the cathode and the substrate to cause the analytes to migrate toward the capture probe on the substrate.

In some implementations, the system can optionally include one or more of the following features. The capture probe may be immobilized on the substrate. The substrate may include a conductive material. The substrate may be coated with a conductive material. The conductive material may include at least one of tin oxide (TO), indium tin oxide (ITO), a transparent conductive oxide (TCO), aluminum doped zinc oxide (AZO), or fluorine doped tin oxide (FTO). The substrate may include an array of substrate regions configured to place capture probes thereon. The method may include arranging a spacer between the substrate and the cathode to contain the buffer between the substrate and the cathode. The buffer may include a permeabilization reagent. The method may include connecting electrical wires from a power supply with the substrate and the cathode, respectively.

Particular embodiments described herein include an electrophoretic system. The system may include a substrate, a substrate cassette, a cathode assembly, and a controller. The substrate may include a plurality of substrate regions including capture probes and one or more biological samples containing analytes. The substrate cassette may be configured to hold the substrate and include a plurality of apertures corresponding to the plurality of substrate regions of the substrate. The plurality of apertures may be configured to define a plurality of buffer chambers on the plurality of substrate regions of the substrate. The cathode assembly may include a plurality of electrode plates. The plurality of electrode plates may be configured to position within the plurality of buffer chambers of the substrate cassette. The controller may be configured to generate electric fields between the plurality of substrate regions and the plurality of electrode plates, respectively, such that the analytes in the biological samples migrate toward the capture probes on the substrate.

In some implementations, the system can optionally include one or more of the following features. The biological samples may be placed in contact with the capture probes on the plurality of substrate regions. The plurality of substrate regions may include a plurality of wells recessed on the substrate. The substrate cassette may include a substrate holder and a gasket. The substrate holder may include a substrate mount for securing the substrate. The gasket may include a plurality of gasket apertures configured to align with the plurality of substrate regions when the substrate is secured by the substrate holder. The plurality of apertures may include the plurality of gasket apertures. The substrate holder may include a plurality of holder apertures configured to align with the plurality of gasket apertures when the substrate is secured by the substrate holder. The plurality of apertures may include the plurality of gasket apertures and the plurality of holder apertures. The system may include a substrate cover configured to be arranged on the substrate cassette and include a plurality of cover apertures configured to align with the plurality of apertures of the substrate cassette. The cathode assembly may be mounted to the substrate cover and the plurality of electrode plates of the cathode assembly may be configured to extend through the plurality of cover apertures into the plurality of apertures of the substrate. The substrate may be coated with a conductive material. The conductive material may include at least one of tin oxide (TO), indium tin oxide (ITO), a transparent conductive oxide (TCO), aluminum doped zinc oxide (AZO), or fluorine doped tin oxide (FTO). The buffer may include a permeabilization reagent. The system may include a power supply, and electrical wires connecting the power supply to the cathode and each of the plurality of substrate regions of the substrate.

Particular embodiments described herein include a method for capturing analytes from a biological sample. The method may include placing biological samples in contact with capture probes on a substrate, the biological sample including analytes; arranging a substrate cassette onto the substrate to align a plurality of apertures of the substrate cassette with a plurality of substrate regions of the substrate and define a plurality of buffer chambers on the plurality of substrate regions; supplying buffers in the plurality of buffer chambers; arranging a cathode to place a plurality of electrode plates of the cathode within the plurality of buffer chambers; and generating electric fields between the plurality of substrate regions and the plurality of electrode plates to cause the analytes in the biological samples to migrate toward the capture probes on the substrate.

In some implementations, the system can optionally include one or more of the following features. The capture probes may be immobilized on the plurality of substrate regions. The biological samples may be placed in contact with the capture probes on the plurality of substrate regions. The plurality of substrate regions may include a plurality of wells recessed on the substrate. The substrate cassette may include a substrate holder and a gasket. The substrate holder may include a substrate mount for securing the substrate. The gasket may include a plurality of gasket apertures configured to align with the plurality of substrate regions when the substrate is secured by the substrate holder. The plurality of apertures may include the plurality of gasket apertures. The substrate holder may include a plurality of holder apertures configured to align with the plurality of gasket apertures when the substrate is secured by the substrate holder. The plurality of apertures may include the plurality of gasket apertures and the plurality of holder apertures. The method may include providing a substrate cover including a plurality of cover apertures and mounting the cathode assembly; and placing the substrate cover onto the substrate cassette such that the plurality of cover apertures of the substrate cover is aligned with the plurality of apertures of the substrate cassette, respectively, and such that the plurality of plates of the cathode assembly extends through the plurality of cover apertures into the plurality of apertures of the substrate. The substrate may be coated with a conductive material. The conductive material may include at least one of tin oxide (TO), indium tin oxide (ITO), a transparent conductive oxide (TCO), aluminum doped zinc oxide (AZO), or fluorine doped tin oxide (FTO).

The devices, system, and techniques described herein may provide one or more of the following advantages. Some embodiments described herein include an electrophoretic system configured to provide appropriate permeabilization of a sample and reduce lateral diffusion that may result from incomplete permeabilization of the sample, thereby increasing the amount of analytes captured and available for detection.

Further, the electrophoretic system described herein can eliminate or reduce additional permeabilization processes. The electrophoretic system can achieve a desired level of spatial resolution without other types of permeabilization or with reduced additional permeabilization. For example, the electrophoretic system can eliminate a need of a permeabilization agent. Alternatively, the electrophoretic system can permit for a reduced amount of permeabilization agent to be used to achieve a desired level of spatial resolution. For example, prior to electrophoresis, a sample can be contacted with a permeabilization agent only for a shorter period of time than a time for complete permeabilization of the sample. Such incomplete permeabilization of the sample can be compensated by the electrophoretic system described herein that eliminates or reduces lateral diffusion of migrating analytes (i.e., analyte diffusion in the transverse direction–orthogonal to the normal direction to the surface of the sample).

Moreover, the electrophoretic system described herein can cause analytes to actively migrate to capture probes by electrophoretic transfer, thereby permitting for the spatial location of the analytes captured by the capture probes on a substrate to be more precise and representative of the spatial location of the analytes in the biological sample than when the analytes are migrated to the capture probes under different environments.

All publications, patents, and patent applications mentioned in this specification are herein incorporated by reference to the same extent as if each individual publication, patent, patent application, or item of information was specifically and individually indicated to be incorporated by reference. To the extent publications, patents, patent applications, and items of information incorporated by reference contradict the disclosure contained in the specification, the specification is intended to supersede and/or take precedence over any such contradictory material.

Where values are described in terms of ranges, it should be understood that the description includes the disclosure of all possible sub-ranges within such ranges, as well as specific numerical values that fall within such ranges irrespective of whether a specific numerical value or specific sub-range is expressly stated.

The term "each," when used in reference to a collection of items, is intended to identify an individual item in the collection but does not necessarily refer to every item in the collection, unless expressly stated otherwise, or unless the context of the usage clearly indicates otherwise.

Various embodiments of the features of this disclosure are described herein. However, it should be understood that such embodiments are provided merely by way of example, and numerous variations, changes, and substitutions can occur to those skilled in the art without departing from the scope of this disclosure. It should also be understood that various alternatives to the specific embodiments described herein are also within the scope of this disclosure.

DESCRIPTION OF DRAWINGS

The following drawings illustrate certain embodiments of the features and advantages of this disclosure. These embodiments are not intended to limit the scope of the appended claims in any manner. Like reference symbols in the drawings indicate like elements.

FIG. **1** schematically illustrates an example electrophoretic system.

FIG. **2** schematically illustrates an example substrate.

FIGS. **3**A-D illustrate example substrates and samples undergoing electrophoretic process.

FIG. **4** is a flowchart of an example process for preparing a sample.

FIG. **5** schematically illustrates an example electrophoretic system.

FIG. **6** schematically illustrates an example substrate.

FIG. **7** shows an example substrate holder with a substrate in an assembled state.

FIG. **8** shows a bottom surface of an example substrate holder including a gasket and receiving a substrate.

FIGS. **9**A-C show A) a top view of an example substrate holder in an open position, B) a side view of a latitudinal side of the substrate holder and C) is a cross sectional view of the substrate holder.

FIGS. **10**A and B illustrate an example placement of a substrate into the substrate holder.

FIG. **11** schematically illustrates an example configuration of a substrate region and a sample undergoing electrophoretic process in each buffer chamber.

FIG. **12** is a flowchart of an example process for preparing a sample.

FIG. **13** schematically illustrates an example system for preparing a sample.

FIG. **14** illustrates another example substrate.

FIGS. **15**A-C are A) a perspective view of an example device including a plate and a substrate holder for heating a substrate, B) top perspective view of the plate, and C) a bottom perspective view of the plate.

FIG. **16** is an exploded view of the substrate holder of FIG. **15**.

FIGS. **17**A-B are A) a top perspective view of a bottom member of the substrate, and B) a bottom perspective view of the bottom member of FIG. **16**.

FIG. **18** is a perspective view of the bottom member of FIG. **17**A coupled to the plate of FIG. **15**.

FIG. **19** is a perspective view of the bottom member of FIG. **17**A coupled to a second plate embodiment.

FIGS. **20**A-B are A) a front perspective view of a fastener for use with the bottom member of FIG. **17** and B) a back perspective view of the fastener.

FIG. **21** is a perspective view of a gasket for use with the substrate holder of FIG. **16**.

FIGS. **22**A-B are a A) top perspective view of a top member of the substrate holder, and B) a bottom perspective view of the top member of FIG. **16**.

FIGS. **23**A-C are A) a perspective view of an example substrate holder, B) an exploded view of the substrate holder, and C) a partial, perspective view of the substrate holder.

FIGS. **24**A-C are A) a top perspective view of the substrate, B) a bottom perspective view of the substrate, and C) a side perspective view of the substrate holder of FIG. **23**A.

FIG. **25** is a perspective view of an example gasket.

DETAILED DESCRIPTION

In general, the present disclosure provides electrophoretic apparatuses, systems, and methods for preparing a sample for a spatial analysis described herein.

Some embodiments include an electrophoretic system for preparing a biological sample, such as a cell or a tissue sample including a cell. The electrophoretic system can be used to permeabilize the sample to allow analytes to be released from the sample (e.g., the cell therein). For example, the sample can be contacted with capture probes attached to a substrate (e.g., a surface of the substrate), and an electric field created by the electrophoretic system can cause analytes to be released from the cell, and effectively migrate toward and bind to the capture probes attached to the substrate. The electrophoretic system described herein can actively direct analytes released from a cell to the capture probes, thereby improving spatial resolution by eliminating or reducing diffusive migration.

In some implementations, a substrate on which a sample is placed can include a conductive material that can be used as an anode in the electrophoretic system described herein. An electrode can be placed in an arrangement in the system such that the electrode is spaced apart from the substrate. The electrode can be used as a cathode in the system. A non-conductive spacer can be arranged between the substrate and the electrode. In some embodiments, one or more conductive materials, electrodes, and/or non-conductive spacers can be used in the system described herein. The substrate, the electrode, and the spacer can be at least partially immersed in a buffer. In some embodiments, the substrate, the electrode, the spacer, or any combination thereof, can be fully immersed in a buffer. The electrophoretic system can apply a voltage between the electrodes (i.e., the substrate as the anode, and the electrode as the cathode) to cause analytes to release from the sample and migrate to capture probes attached to the substrate.

In some implementations, a variety of substrate cassettes for analyzing multiple samples can be used with the electrophoresis system provided herein. For example, a substrate cassette can be configured to engage a substrate that has multiple substrate regions. The substrate cassette can include a plurality of apertures corresponding to the substrate regions of the substrate, respectively. The plurality of apertures can be used as a plurality of chambers for receiving buffers, respectively. The substrate can be made of a conductive material and used as a common anode for the plurality of chambers. A cathode can be configured to include a plurality of electrode plates that can extend into the chambers at least partially filled with buffers, respectively. The electrophoretic system can apply a voltage between the substrate (as the anode) and the cathode (including the electrode plates) to cause analytes to release from the samples in each of the substrate regions, and migrate to a capture probe provided at each of the substrate regions of the substrate.

Referring to FIGS. **1**-**4**, an example electrophoretic system **3000** is described for preparing a sample. FIG. **1** schematically illustrates an example configuration of the electrophoretic system **3000**. In some implementations, the electrophoretic system **3000** can be used to provide electrophoretic permeabilization of a sample, and/or actively cause analytes in the sample to migrate to capture probes on a substrate. In some embodiments, the electrophoretic system **3000** can be used to enhance electrophoretic permeabilization of a sample by actively directing analytes in the sample with desired directionality. For example, electrophoretic permeabilization by the system **3000** can result in higher analyte capture events (by, e.g., driving more analytes to the capture probes) and better spatial fidelity of captured analytes (e.g., on a feature array) than random diffusion onto matched substrates without the application of an electric (e.g., increases resolution of spatial analyte detection).

In some implementations, the electrophoretic system **3000** can include a substrate **3002** as a first electrode, a second electrode **3004**, an electrophoretic container **3006**, and a control system **3008**. The system **3000** can further include a spacer **3010**.

Referring to FIG. **2**, the substrate **3002** is configured to receive a sample **3012** that contains analytes. The sample **3012** includes a biological sample, such as a cell or a tissue including a cell. The substrate **3002** can include a substrate region **3016** for receiving the sample **3012** thereon. In some implementations, the substrate **3002** can place a plurality of capture probes **3018** on the substrate region **3016**. The capture probes **3018** can be placed on the substrate region **3016** in various manners, such as a variety of ways generally described herein. For example, the capture probes **3018** can be directly attached to a feature that is on an array. Alternatively or in addition, the capture probes **3018** can be immobilized on the substrate region **3016** of the substrate **3002**. The sample **3012** can be prepared on the substrate **3002** in various ways generally described herein.

In some implementations, the substrate **3002** is configured to be used as a first electrode in the electrophoretic system **3000**. For example, the substrate **3002** can be used as an anode. In another example, the substrate **3002** can be used as a cathode.

The substrate **3002** can be configured as a conductive substrate described generally herein. For example, the substrate **3002** can include one or more conductive materials that permit for the substrate **3002** to function as an electrode (e.g., the anode). Examples of such a conductive material include tin oxide (TO), indium tin oxide (ITO), a transparent conductive oxide (TCO), aluminum doped zinc oxide (AZO), fluorine doped tin oxide (FTO), and any combination thereof. Alternatively or in addition, other materials may be used to provide desired conductivity to the substrate **3002**. In some implementations, the substrate **3002** can be coated with the conductive material. For example, the substrate **3002** can include a conductive coating on the surface thereof, and the sample **3012** is provided on the coating of the substrate **3002**.

Although the substrate **3002** is illustrated to include a single substrate region **3016** in FIG. **2**, other implementations of the substrate **3002** can include a plurality of substrate regions that are configured to place capture probes and/or samples thereon, respectively.

In the illustrated example of FIG. **1**, the substrate **3002** is used as the anode, and the second electrode **3004** is configured as a cathode. In this example, therefore, the second electrode **3004** can also be referred to as the cathode **3004**. In some implementations, the cathode **3004** can include a conductive plate, one or more pins, or other suitable configurations. In other implementations, the cathode **3004** can include a conductive substrate that is similar to the substrate **3002**. In this configuration, in some implementations, the cathode **3004** does not include capture probes, agents, solutions, or other substances or materials that may interact with the sample placed on the substrate **3002**.

The substrate **3002** and the cathode **3004** can be arranged within the electrophoretic container **3006**. The electrophoretic container **3006** can provide a buffer chamber **3022** between the substrate **3002** and the cathode **3004**. The buffer chamber **3022** is configured to contain a buffer **3024**. In some implementations, the substrate **3002** and the cathode **3004** can be fully immersed into the buffer **3024**. In alternative implementations, either or both of the substrate **3002** and the cathode **3004** can be partially inserted into the buffer **3024** contained in the electrophoretic container **3006**.

The buffer **3024** can be of various types. In some implementations, the buffer **3024** includes a permeabilization reagent. In some implementations, the buffer **3024** does not include a permeabilization reagent. The buffer **3024** is contained in the buffer chamber **3022** throughout the electrophoretic process.

The spacer **3010** can be disposed between the substrate **3002** and the cathode **3004** to space them apart at a distance D. The spacer **3010** is made of non-conductive material, such as plastic, glass, porcelain, rubber, etc. The distance D can be determined to provide a desired level of spatial resolution based on several factors, such as the strength and/or duration of electric field generated between the substrate **3002** and the cathode **3004**, and other parameters described herein. The spacer **3010** can define at least part of the buffer chamber **3022** between the substrate **3002** and the cathode **3004**.

The controller **3008** operates to generate an electric field (−E) between the substrate **3002** and the cathode **3004**. As illustrated in FIG. **3**, the analytes **3014** in the sample **3012** can migrate toward the capture probes **3018** under the electric field (−E). The controller **3008** can operate to apply a voltage between the substrate **3002** and the cathode **3004** using a power supply **3020**. The power supply **3020** can include a high voltage power supply. The controller **3008** can be electrically connected to the substrate **3002** and the cathode **3004** using electrical wires **3023**.

FIGS. **3**A-D illustrate examples of configurations of a substrate and a sample undergoing electrophoretic process using the electrophoretic system **3000**. The application of electric field (−E) causes the analytes **3014** to move towards the capture probes **3018** in the direction of the arrow shown. In some implementations, the analytes **3014** include a protein or a nucleic acid. In some embodiments, the analytes **3014** are negatively charged proteins or nucleic acids. In some embodiments, the analytes **3014** include a positively charged protein or a nucleic acid. In some embodiments, the analytes **3014** includes a negatively charged transcript. For example, the analytes **3014** include a polyA transcript. In some embodiments, the capture probes **3018** are attached to the substrate **3002**. In some embodiments, the capture probes **3018** can be attached on a feature of an array array. In some embodiments, the analytes **3014** move towards the capture probes **3018** for a distance (h). In some embodiments, the buffer **3024** (e.g., including a permeabilization reagent) can be in contact with the sample **3012**, the substrate **3002**, the cathodes **3004**, or any combination thereof. The buffer **3024** can include any of the permeabilization reagents disclosed above including but not limited to a permeabilization reagent, a permeabilization buffer, a permeabilization enzyme, a buffer without a permeabilization reagent, a permeabilization gel, and a permeabilization solution.

FIG. **3**B shows another example configuration in which the sample **3012** is on a first substrate **3106**, and there is a gap **3114** in between the sample **3012** and a coating **3110** on the surface of a second substrate **3108**. The coating **3110** can be a conductive coating as described herein. For this embodiment, applied electrophoretic charge causes the analytes **3102** to migrate from the biological sample **3012** on the first substrate, through the buffer **3112** and across the gap **3114** to the capture probes **3104** disposed on the second substrate **3108**. FIG. **3**C shows another example configuration in which sample **3012** is on coating **3110** of the second substrate **3108**, but there is no gap present in between the sample **3012** and the first substrate **3106**. FIG. **3**D shows another example configuration similar to that of FIG. **3**B in which sample **3012** is on the first substrate **3106**, but there is no or minimal gap present between the sample **3012** and the second substrate **3108** upon which is located the capture probes **3104**.

FIG. **4** is a flowchart of an example process **3200** for preparing a sample for use in the electrophoretic systems described herein. In some implementations, the process **3200** includes providing a substrate including capture probes (**3202**), and placing a sample in contact with the the substrate (**3204**). The capture probes can be attached to the substrate in various ways generally described herein. The sample can be placed on the substrate in various ways generally described herein. As described herein, the substrate can be configured and used as an electrophoretic electrode (also referred to herein as a first electrode). In some implementations, the substrate can be configured as a conductive substrate as described herein, such as by including a conductive material in the substrate or providing a conductive coating on an upper or lower surface of the substrate. In some implementations, the substrate can be used as an anode. In alternative implementations, the substrate can be used as a cathode.

The process **3200** can further include arranging a spacer on or above the substrate (**3206**). When a second electrode is arranged as described below, the spacer can be arranged between the second electrode and the first electrode (e.g., the substrate). As described herein, the spacer can be made of a non-conductive material and used to provide a buffer chamber between the first and second electrodes.

The process **3200** can include arranging a second electrode relative to the first electrode (e.g., the substrate) at a distance (**3208**). The second electrode can be used as a cathode when the substrate is used as an anode. Alternatively, the second electrode can be used as an anode when the substrate is used as a cathode. The second electrode can be made of various configurations. For example, the second electrode can include a conductive plate. Alternatively, the second electrode can be configured as a conductive substrate that is configured similarly to the first electrode (e.g., the substrate).

The process **3200** can include providing a buffer between the first electrode and the second electrode (**3210**). The buffer can be contained in the buffer chamber that is provided by the spacer and used to at least partially immerse the first electrode (e.g., the substrate), the second electrode, or both. In some implementations, the buffer can include a permeabilization reagent. Other buffers as described generally herein can be used in other implementations. In some embodiments, both electrodes are immersed in a buffer.

The process **3200** can include generating an electric field between the first electrode (e.g., the substrate) and the second electrode (**3212**). Under the electric field, analytes included in the sample can migrate toward the capture probe on the substrate, wherein the analytes can hybridize to the captures probes. In some implementations, the electric field is generated by applying a voltage between the first and second electrodes, using a power supply electrically connected to the first and second electrodes. For example, the process **3200** can include connecting electrical wires from the power supply with the first electrode (e.g., the substrate) and the second electrode.

Referring to FIGS. **5**-**12**, another example electrophoretic system **4000** is described. FIG. **5** schematically illustrates an example configuration of the electrophoretic system **4000**. In some implementations, the electrophoretic system **4000** can be used to provide electrophoretic permeabilization of a plurality of samples, and/or actively cause analytes in each sample to migrate to capture probes on a substrate. For example, electrophoretic permeabilization by the system **4000** can result in more analytes being captured by capture probes and better spatial fidelity of captured analytes (e.g., on a feature array) than random diffusion onto substrates without the application of an electric field.

In some implementations, the electrophoretic system **4000** can include a substrate **4002** as a first electrode, a second electrode **4004**, a control system **4008**, and a substrate cassette **4010**.

Referring to FIG. **6**, the substrate **4002** is configured to receive a plurality of samples **4012** that contain analytes. The samples **4012** can include biological samples **4012**, such as a cell or a tissue including a cell. The substrate **4002** can include a plurality of substrate regions **4016** for receiving the sample **4012** thereon. In some implementations, the substrate **4002** a plurality of capture probes **4018** on each substrate region **4016**. The capture probes **4018** can be placed on the substrate region **4016** in various manners, such as a variety of ways generally described herein. For example, the capture probes **4018** can be directly attached to a feature on an array. Alternatively or in addition, the capture probes **4018** can be immobilized on the substrate region **4016** of the substrate **4002**. The sample **4012** can be prepared on the substrate **4002** in various ways generally described herein.

In some implementations, the substrate **4002** is configured to be used as a first electrode in the electrophoretic system **4000**. For example, the substrate **4002** can be used as an anode. In another example, the substrate **4002** can be used as a cathode.

The substrate **4002** can be configured as a conductive substrate as described generally herein. For example, the substrate **4002** can include a conductive material that permits for the substrate **4002** to function as an electrode (e.g., the anode). Examples of such a conductive material include tin oxide (TO), indium tin oxide (ITO), a transparent conductive oxide (TCO), aluminum doped zinc oxide (AZO), fluorine doped tin oxide (FTO), and any combination thereof. Alternatively or in addition, other materials may be used to provide conductivity to the substrate **4002**. In some implementations, the substrate **4002** can be coated with the conductive material. For example, the substrate **4002** can include a conductive coating on the surface thereof, and the sample **4012** is provided on the coating of the substrate **4002**. Other examples of the substrate **4002** are further described below, for example with reference to FIG. **14**.

Referring back to FIG. **5**, the substrate cassette **4010** is configured to accommodate the substrate **4002**. For example, the substrate cassette **4010** can be configured to immovably mount and hold the substrate **4002**. The substrate cassette **4010** can include a plurality of apertures **4006**. The plurality of apertures **4006** can be positioned in the substrate cassette **4110** so as to correspond to the plurality of substrate regions **4016** of the substrate **4002** when the substrate cassette **4010** mounts the substrate **4002**. The plurality of apertures **4006** can define a plurality of buffer chambers **4022** on the plurality of substrate regions **4106** of the substrate **4002**, respectively. The substrate cassette **4010** can be made of non-conductive material, such as plastic, glass, porcelain, rubber, etc.

Referring to FIGS. **7-10**, an example substrate cassette **4190** is illustrated. The substrate cassette **4190** can be used to implement the substrate cassette **4010** in FIG. **5**. The substrate cassette **4190** can include a substrate holder **4200** configured to hold a substrate.

FIG. **7** shows an example substrate holder **4200** with a substrate in an assembled state. This embodiment of the shown substrate holder **4200** can advantageously provide a single-piece component that can be arranged in an open configuration or closed configuration, when desired. In particular, FIG. **7** shows a top surface **4293** of the substrate holder **4200** in a closed position. The substrate holder **4200** includes a plurality of apertures **4256**, which can be used to implement the plurality of apertures **4006** of the substrate cassette **4010** of FIG. **5**.

The substrate holder **4200** can include a substrate loading mechanism for loading and holding the substrate. For example, the substrate loading mechanism can include a first tab **4250***a* and a second tab **4250***b*. In some embodiments, any type of fastener or engagement feature that allows releasable engagement can be used instead of the first and second tabs **4250***a* and **4250***b*, such as, for example, screws and press fit type connectors. In some embodiments, the substrate holder **4200** includes 5 tabs or less (e.g., 4 tabs or less, 3 tabs or less, 2 tabs or less, or 1 tab). In some embodiments, the substrate holder **4200** is a single molded unit. Any suitable plastic or polymer can be used as a suitable molding material.

In some embodiments, the substrate holder **4200** includes a substrate mount that has a first surface and a second surface, where the second surface of the substrate mount is configured to mount a substrate for receiving a sample. The substrate holder can include a first portion configured to receive a gasket. The first portion can include a plurality of ribs extending from a surface of the substrate holder. The substrate holder can include a second portion configured to receive a substrate. The first and second portions can be coupled together by a hinge. The first portion can be configured to fold over the second portion to secure the substrate between the first and second portions. In some embodiments, the substrate is a glass slide. In some embodiments, the substrate holder comprises a gasket disposed between the first portion and the second portion of the substrate holder. In some embodiments, the first portion of the substrate holder includes a releasable engagement mechanism configured to secure the first portion to the second portion when the substrate holder is in the closed state. In some embodiments, the first surface of the substrate engages with at least one of the plurality of ribs extending from a surface of the substrate holder. In some embodiments, the second portion defines a recessed cavity formed in the substrate holder configured to receive the substrate. In some embodiments, the second portion can define a cavity configured to receive the substrate.

FIG. **8** shows the bottom surface **4297** of the top component of the substrate holder **4200** of the substrate cassette **4190**, which includes a gasket **4254** and receiving a slide **4252** (e.g., the substrate). The substrate holder **4200** has longitudinal sides **4205** and latitudinal sides **4207**. First and second tabs **4250***a* and **4250***b* (FIG. **9**), respectively, can protrude from a longitudinal side **4205** of the substrate holder **4200**. In some embodiments, the substrate holder **4200** is a single molded unit that includes the gasket **4254**. That is, the substrate holder **4200** and the gasket **4254** are one part. In some embodiments, the substrate holder **4200** is overmolded with the gasket **4254**. For example, the substrate holder **4200** is a first injection molded plastic part with a second part (e.g., a pliable material) molded onto it to create the gasket **4254**. In some embodiments, the pliable material is an elastomer. In some embodiments, the pliable material is silicone rubber. In some embodiments, the gasket **4254** is a separate part that is not molded with the substrate holder **4200**.

FIG. **9**A shows a top view of the substrate holder **4200** of the substrate cassette **4190** in an open position. The opening and closing mechanism of the substrate holder **4200** is a hinged mechanism. A bottom component **4362** of the substrate holder **4200** can be hinged to a top component **4364** of the substrate holder **4200** via a hinge **4360**. In some embodiments, the hinge **4360** can be a living hinge. In some embodiments, the substrate holder **4200** includes 10 hinges or less (e.g., 9 hinges or less, 8 hinges or less, 7 hinges or less, 6 hinges or less, 5 hinges or less, 4 hinges or less, 3 hinges or less, 2 hinges or less, or 1 hinge). Non-limiting examples of hinges that the substrate holder **4200** can include, include a straight or flat living hinge, a butterfly living hinge, a child safe hinge, a double living hinge, and a triple living hinge.

The substrate holder **4200** further includes one or more engagement features, such as a first notch **4358***a* and a second notch **4358***b*. The first and second notches **4358***a* and **4358***b* can engage the first and second tabs **4250***a* and **4250***b*, respectively, when pressed together. The first and second notches **4358***a* and **4358***b* can protrude from a longitudinal side **4205** of the top component **4364** of the substrate holder **4200**. In some embodiments, the substrate holder **4200** includes three, four, five, six, seven, eight, nine, ten or more notches. In some embodiments, the notches protrude from a latitudinal side **4207** of the substrate holder **4200**. In some embodiments, the first and second notches **4358***a* and **4358***b* are rigid and do not flex when engaging the first and second tabs **4250***a* and **4250***b*, respectively. In some embodiments, the first and second notches **4358***a* and **4358***b*, respectively, can be flexible.

FIG. **9**B shows a side view of a latitudinal side **4207** of the substrate cassette **4190**. The first and second notches **4358***a* and **4358***b* can project upward and include a notch ledge **4366** that engages a tab ledge **4368**, as shown in FIG. **9**C. Alternatively, in some embodiments, the substrate holder **4200** includes a snap fit locking mechanism for releasably receiving and releasably securing the slide **4252**. Non-limiting examples of other types of fasteners to be used in locking mechanisms of the substrate holder **4200** include a catch, a projection, a male connector, and a female connector.

FIGS. **10**A and B illustrate the placement of the slide **4252** into the substrate holder **4200** of the substrate cassette **4190**. In some embodiments, the slide **4252** can be "loaded" or placed onto an inner rim or an inner edge of the bottom component **4362** of substrate holder **4200**. Once loaded, the top component **4364** is closed by pressing the first notch **4358***a* and the second notch **4358***b* against the first and second tabs **4250***a* and **4250***b*, respectively, thereby forming a tight seal with the slide **4252**. In some embodiments, the slide does not have to be tilted under tabs **4250** or any other tabs. In some embodiments, the substrate holder **4200** includes one or more tabs to help load the slide onto the inner rim or inner edge of the bottom component **4362** of substrate holder **4200**.

A variety of other configurations of substrate holder can be used to implement the substrate cassette **4010** in FIG. **5**. Examples of such other configurations are further illustrated below, for example with reference to FIG. **15**.

Referring back to FIG. **5**, the substrate **4002** is used as the anode, and the second electrode **4004** is configured as a cathode. In this example, therefore, the second electrode **4004** can also be referred to as the cathode **4004**. In some implementations, the cathode **4004** can include a conductive plate, one or more pins, or other suitable configurations. As illustrated, for example, the cathode **4004** can include a plurality of electrode plates **4104**. The electrode plates **4104** are configured to be positioned within the plurality of buffer chambers **4022** of the substrate cassette **4010**, respectively.

The samples **4012** on the substrate regions **4016** of the substrate **4002**, respectively, can be arranged within the buffer chambers **4022** that are defined by the apertures **4006** of the substrate cassette **4010**. The buffer chambers **4022** can contain buffers **4024** therein, so that the samples **4012** are fully immersed into the buffers **4024** in the buffer chambers **4022**, respectively. Further, as illustrated in FIGS. **5** and **7**, the electrode plates **4104** of the cathode **4004** can be arranged within the buffer chambers **4022**, respectively. Each of the electrode plates **4104** can be immersed into a buffer **4024** contained in the buffer chamber **4022**. The electrode plates **4104** are arranged to be spaced apart from the corresponding substrate regions **4016** of the substrate **4002**.

The buffer **4024** can be of various types. In some implementations, the buffer **4024** includes a permeabilization reagent. Alternatively or in addition, other type of buffers as generally described herein can be used. The buffer **4024** is contained in each of the buffer chambers **4022** throughout the electrophoretic process.

The controller **4008** operates to generate an electric field (−E) between the substrate (anode) **4002** and the cathode **4004**. For example, the controller **4008** operates to generate an electric field between the substrate regions **4016** of the substrate **4002** and the electrode plates **4104** of the cathode **4004** that correspond to the substrate regions **4016**, respectively. Thus, the electric field is generated through each of the buffer chambers **4022**. In some implementations, the same electric field is generated for all of the buffer chambers **4022**. In other implementations, different electric fields are generated for at least two of the buffer chambers **4022**.

In some implementations, the controller **4008** can operate to apply a voltage between the entire substrate **4002** and the entire cathode **4004** using a power supply **4020**. In other implementations, the controller **4008** can operate to apply voltages between the substrate regions **4016** of the substrate **4002** and the corresponding electrode plates **4104** of the cathode **4004**, respectively. The power supply **4020** can include a high voltage power supply. The controller **4008** can be electrically connected to the substrate **4002** and the cathode **4004** using electrical wires **4023**.

FIG. **11** schematically illustrates an example configuration of a substrate region and a sample undergoing electrophoretic process in each buffer chamber of the electrophoretic system **4000**. In each buffer chamber **4022**, the analytes **4014** in the sample **4012** can migrate toward the capture probes **4018** under the electric field (−E).

The application of electric field (−E) can cause the analytes **4014** (e.g., negatively charged analytes) to move towards the capture probes **4018** (e.g., positively charged analytes) in the direction of the arrow shown. In some implementations, the analytes **4014** include a protein or a nucleic acid. In some embodiments, the analytes **4014** are negatively charged proteins or nucleic acids. In some embodiments, the analytes **4014** include a positively charged protein or a nucleic acid. In some embodiments, the analytes **4014** includes a negatively charged transcript. For example, the analytes **4014** include a polyA transcript. In some embodiments, the capture probes **4018** are affixed on the substrate **4002** at the substrate regions **4016** of the substrate **4002**. In some embodiments, the capture probes **4018** are location at a feature on the array, or can be replaced at a feature on the array. In some embodiments, the analytes **4014** move towards the capture probes **4018** for a distance (h). In some embodiments, the buffer **4024** (e.g., comprising a permeabilization reagent) can be in contact with the sample **4012**, the substrate **4002** (e.g., the substrate region **4016** thereof), the cathode **4004** (e.g., the electrode plates **4104** thereof), or any combination thereof. The buffer **4024** can include any of the permeabilization reagents disclosed including but not limited to a permeabilization reagent such as a permeabilization enzyme, a permeabilization buffer, a buffer without a permeabilization reagent, a permeabilization gel, and a permeabilization solution.

In other implementations, the electrophoretic system **4000** can be configured to provide different configurations in the buffer chambers **4024**, using, for example, those described in FIG. **3** above.

FIG. **12** is a flowchart of an example process **4500** for preparing a sample. In some implementations, the process **4500** includes providing a substrate including a plurality of substrate regions (**4502**). Each substrate region of the substrate can include a plurality of capture probes. The process **4500** can include placing multiple samples in contact with the capture probes on the substrate regions of the substrate (**4504**), for example, when there are two substrate regions one sample is contacted to each region, when there are three regions, there are three samples (one on each region), etc. The capture probes can be attached to the substrate regions in various ways described herein. The sample can be placed on the substrate region in various ways described herein.

The substrate can be configured and used as an electrophoretic electrode (also referred to herein as a first electrode). In some implementations, the substrate can be configured as a conductive substrate as described herein, such as by including a conductive material in the substrate or providing a conductive coating on an upper or lower surface of the substrate. In some implementations, the substrate can be used as an anode. In alternative implementations, the substrate can be used as a cathode.

Alternatively, the substrate can be configured and used such that each substrate region operates as an electrophoretic electrode (also referred to herein as a first electrode), while the other portion (e.g., at least a portion around each conductive substrate region) is non-conductive. In some implementations, each substrate region can be configured as a conductive region, such as by including a conductive material in the substrate region or providing a conductive coating on an upper or lower surface of the substrate region. In some implementations, each substrate region can be used as an anode. In alternative implementations, each substrate region can be used as a cathode.

The process **4500** can further include arranging a substrate cassette on or above the substrate (**4506**). The substrate cassette can be arranged such that a plurality of apertures of the substrate cassette are aligned with the substrate regions of the substrate, respectively, thereby defining a plurality of buffer chambers on the plurality of substrate regions. As described herein, the substrate cassette can be at least partially made of a non-conductive material and used to provide the buffer chambers between the first and second electrodes.

The process **4500** can include arranging a second electrode relative to the first electrode (e.g., the substrate or each substrate region) at a distance (**4508**). The second electrode can be used as a cathode when the substrate or each substrate region is used as an anode. Alternatively, the second electrode can be used as an anode when the substrate or each substrate region is used as a cathode. The second electrode can be made of various configurations. For example, the second electrode can include a plurality of electrode plates configured to be placed within the buffer chambers, respectively.

The process **4500** can include providing a buffer between the first electrode (e.g., the substrate or each substrate region) and the second electrode (e.g., each electrode plate) (**4510**). The buffer can be contained in each buffer chamber that is provided by the substrate cassette and used to at least partially immerse the first electrode (e.g., the substrate or each substrate region), the second electrode (e.g., each electrode plate), or both. In some implementations, the buffer can include a permeabilization reagent.

The process **4500** can include generating an electric field between the first electrode (e.g., the substrate or each substrate region) and the second electrode (e.g., each electrode plate) (**4512**). Under the electric field, analytes included in the sample can migrate toward the capture probes on each substrate region, for example in order to hybridize (e.g., captured) to the capture probe. In some implementations, the electric field is generated by applying a voltage between the first and second electrodes, using a power supply electrically connected to the first and second electrodes. For example, the process **4500** can include connecting electrical wires from the power supply with the first electrode (e.g., the substrate or each substrate region) and the second electrode (e.g., each electrode plate).

FIG. **13** schematically illustrates an example system **4700** for preparing a sample. The system **4700** can include a sample preparation instrument **4702**, a substrate **4802** as a first electrode, a second electrode **4804**, a control system **4808**, and a substrate cassette **4810**.

The sample preparation instrument **4702** is configured to support the substrate cassette **4810** with the substrate **4802**, and provide the second electrode **4804** to generate an electric field between the substrate **4802** and the second electrode **4804**. In some implementations, the sample preparation instrument **4702** includes a body **4704** and a lid **4706** configured to be place over the body **4704**. For example, the lid **4706** is hingedly connected to the body **4704**. Alternatively, the lid **4706** can be coupled to the body **4704** in other configurations. In yet alternative implementations, the lid **4706** is configured to be separate from the body **4704**, or detachably attached to the body **4704**.

In some implementations, the body **4704** provides a supporting surface **4708** configured to support the substrate cassette **4810** including the substrate **4802**. The body **4704** further provides an electrical connector **4710** for electrically contacting the substrate **4802** so that the substrate **4802** can be used as an electrode (e.g., an anode) for electrophoresis. The substrate **4802** and the substrate cassette **4810** can be configured identically or similarly to the substrate **4002** and the substrate **4010**, respectively. For example, the substrate **4802** includes a plurality of substrate regions that include capture probes configured to place samples thereon. The substrate cassette **4810** includes a plurality of apertures that correspond to the plurality of substrate regions and define a plurality of buffer chambers.

The lid **4706** can be configured to mount the second electrode **4804** (e.g., a cathode). The second electrode **4804** can be configured similarly to the second electrode **4004**. For example, the second electrode **4804** can include a plurality of electrode plates **4814** corresponding to the buffer chambers on the substrate **4802**. The second electrode **4804** is arranged such that, when the lid **4706** is closed onto the body **4704**, or comes close to the body **4704**, the electrode plates **4814** of the second electrode **4004** are at least partially inserted into the buffer chambers of the substrate cassette **4810**.

The sample preparation instrument **4702** can include the control system **4808** that is identical or similar to the controller **4008**. For example, the control system **4808** can be housed in the body **4704**. Alternatively, at least part of the control system **4808** can be housed in or mounted to another part of the instrument, such as the lid **4706**. Alternatively, the control system **4808** can be at least partially configured as a separate apparatus from the instrument **4702** and electrically connected to the instrument **4702**. The control system **4808** can be electrically connected to the substrate **4802** (or each substrate region thereof) and the second electrode **4804** (or each electrode plate thereof) using electrical wires **4822**.

The sample preparation instrument **4702** can further include a power supply **4820** configured to apply a voltage between the substrate **4802** (or each substrate region thereof) and the second electrode **4804** (or each electrode plate thereof). The power supply **4820** can be housed in the body **4704**. Alternatively, the power supply **4820** can be housed in or mounted to another part of the instrument, such as the lid **4706**. Alternatively, the power supply **4820** can be provided separately from the instrument **4702**.

In some implementations, the sample preparation instrument **4702** can be configured to automatically start electrophoresis when the lid **4706** is closed over the body **4704**, or lowered to a predetermined position (or angle) over the samples, and stop electrophoresis when the lid **4706** returns to be opened or raised from the predetermined position. Alternatively, the sample preparation instrument **4702** can provide a user interface **4720** (e.g., a button, switch, etc.) to receive a manual input of starting or stopping the electrophoretic process. The user interface **4720** can include an output device, such as a display, lamps, etc., configured to output operating parameters of the instrument **4702** (e.g., a voltage being applied, a duration of such application, etc.) or other information associated with the system **4700**. The user interface **4720** can further include an input device, such as physical or virtual buttons, switches, keypads, etc., configured to receive a user input of adjusting the operating parameters of the instrument **4702** or other information associated with the system **4700**.

Referring to FIG. **14**, another example substrate **5000** is described. The substrate **5000** can be used to implement the substrates **4002**, and **4802** described herein. The substrate **5000** has a surface **5501** that includes substrate regions **5002***a-h*. The substrate **5000** can include a first substrate region **5002***a*, a second substrate region **5002***b*, a third substrate region **5002***c*, a fourth substrate region **5002***d*, a fifth substrate region **5002***e*, a sixth substrate region **5002***f*, a seventh substrate region **5002***g*, and an eighth substrate region **5002***h*. In some embodiments, substrate **5000** can have less than eight substrate regions or more than eight substrate regions. Each substrate region can be enclosed within a defined perimeter of a frame **5006**, for example a frame comprising fiducial markers.

In some implementations, the eight substrate regions **5002***a-h* can be positioned at the center of the substrate **5000**. For example, the eight substrate regions **5002***a-h* can be centered on the substrate **5000** such that a first distance extending from an outer edge of a frame **5006** of one of the first substrate region **5002***a*, second substrate region **5002***b*, third substrate region **5002***c*, or fourth substrate region **5002***d* to a longitudinal edge (i.e., along length l) of the substrate **5000** is substantially the same to a second distance extending from an outer edge of a frame **5006** of one of the fifth substrate region **5002***e*, sixth substrate region **5002***f*, seventh substrate region **5002***g*, or eighth substrate region **5002***h* to a longitudinal edge (i.e., along length l) of substrate **5000**. For example, the substrate arrays can be positioned about 28.5 mm from the top edge **5032** of substrate, about 11.25 mm from the bottom edge **5034** of substrate, and about 3.5 mm from the left edge **5036** and right edge **5038** of substrate, when the substrate is oriented vertically as is shown in FIG. **14**.

In some embodiments, the eight substrate regions **5002***a-h* can be centered on the substrate **5000** such that a first distance extending from an outer edge of a frame **5006** of one of the first substrate region **5002***a*, second substrate region **5002***b*, third substrate region **5002***c*, or fourth substrate region **5002***d* to a latitudinal edge (i.e., along width w) of the substrate **5000** is substantially the same to a second distance extending from an outer edge of a frame **5006** of one of the fifth substrate region **5002***e*, sixth substrate region **5002***f*, seventh substrate region **5002***g*, or eighth substrate region **5002***h* to a longitudinal edge (i.e., along width w) of the substrate **5000**.

In some embodiments, the substrate **5000** can be rectangular in shape. In some embodiments, the substrate **5000** has a length l of about 100 mm to about 10 mm (e.g., 90 mm or less, 85 mm or less, 90 mm or less, 75 mm or less, 70 mm or less, 65 mm or less, 60 mm or less, 55 mm or less, 50 mm or less, 45 mm or less, 40 mm or less, 35 mm or less, 30 mm or less, 25 mm or less, 20 mm or less, 15 mm or less). In some embodiments, the substrate **5000** has a width w of about 100 mm to about 10 mm (e.g., 90 mm or less, 85 mm or less, 90 mm or less, 75 mm or less, 70 mm or less, 65 mm or less, 60 mm or less, 55 mm or less, 50 mm or less, 45 mm or less, 40 mm or less, 35 mm or less, 30 mm or less, 25 mm or less, 20 mm or less, 15 mm or less). In some embodiments, substrates of the disclosure can be square or circular in shape. In some embodiments, substrates of the disclosure can be rectangular, triangular, hexagonal, octagonal, pentagonal, or any other suitable two-dimensional, geometric shape.

In some embodiments, the substrate **5000** includes 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 or more substrate arrays. The substrate **5000** includes four substrate arrays **5004**, as shown in FIG. **14**. The substrate **5000** can include a substrate array **5004** arranged within the frame **5006**. For example, the substrate array **5004** can be positioned at the center of a frame **5006**. In some embodiments, the substrate arrays **5004** can be the same (e.g., can have a same pattern). In some embodiments, the substrate arrays **5004** can be the different (e.g., can have a different pattern). The four substrate arrays **5004**, shown in FIG. **14**, can be repeated and arranged in a checked pattern. That is, substrate array **5004** is positioned at the center of the first substrate region **5002***a*, the third substrate region **5002***c*, the sixth substrate region **5002***f*, and the eighth substrate region **5002***h*. In some embodiments, the checked pattern is meant to occupy as much of the surface of the substrate to ensure consistency of image acquisition when scanning different regions of the substrate.

In some embodiments, the substrate regions **5002***a-h* can be arranged vertically in two columns, as shown in FIG. **14**. In some embodiments, the substrate regions are arranged on the substrate in one column. In some embodiments, the substrate regions are arranged on the substrate in 2, 3, 4, 5, 6, 7, 8, 9, 10, or more columns. In some embodiments, the substrate arrays **5004** can be positioned adjacent to each other. For example, in some embodiments, the substrate arrays **5004** can be arranged vertically in a column (e.g., positioned on the first four substrate regions **5002***a*-**2402***d*). Alternatively, in other embodiments, the substrate arrays **5004** can be positioned on the substrate regions **5002***a*, **5002***b*, **5002***e*, and **5002***f*. In some embodiments, the substrate arrays **5004** can be positioned on the substrate regions **5002***e*, **5002***f*, **5002***g*, and **5002***h*. In some embodiments, the substrate arrays **5004** can be positioned on the substrate regions **5002***c*, **5002***d*, **5002***e*, and **5002***h*. In some embodiments, the substrate arrays **5004** can be positioned on the substrate regions **5002***e*, **5002***b*, **5002***g*, and **5002***d*. In some embodiments, the substrate arrays **5004** are not arranged in a particular pattern. In some embodiments, the substrate arrays **5004** vary individually in size. For example, in some embodiments, a first substrate array may have a greater size than the size of a second substrate array. In some embodiments, a first substrate array may have a smaller size than the size of a second substrate array. In some embodiments, the distance between substrate arrays may vary and be different. For example, in some embodiments, the distance between a first substrate array and a second substrate array may be different than the distance between a third substrate array and a fourth substrate array. In various embodiments, the substrate **5000** can have eight substrate arrays **5004** or less (e.g., 7 substrate arrays or less, 6 substrate arrays or less, 5 substrate arrays or less, 4 substrate arrays or less, 3 substrate arrays or less, 2 substrate arrays or less). In some embodiments, the substrate arrays are arranged on the substrate in one column. In some embodiments, the substrate arrays are arranged on the substrate in 2, 3, 4, 5, 6, 7, 8, 9, 10, or more columns.

Referring to FIGS. **15-22**, an example substrate cassette is described. In some implementations, an example system **6102** is configured to heat a substrate and includes a plate **6110** and a substrate holder **6150**. The substrate holder **6150** can be used to replace the substrate cassette **4010**, **4810** described herein. The plate **6110** can be configured to be received by a heating device (e.g., a thermocycler) and provide heat transfer between the heating device and the substrate holder **6150**. The substrate holder **6150** holds one or more substrates (such as one or more microscope slides), and can removably couple to the plate **6110** to facilitate heat transfer from the plate **6110** to the one or more substrates.

Referring to FIGS. **15**B and **15**C, the plate **6110** can include a platform **6212** with a first surface **6314** and a second surface **6216**. The plate **6110** can further include a plurality of members **6318** extending from the first surface **6314**. In some embodiments, the plate **6110** can include one or more support members **6220**.

The plate **6110** can be generally formed of thermally conductive material to facilitate heat transfer between a heating device and the substrate holder **6150**. In some embodiments, the plate **6110** can be made of a metal such as (but not limited to) aluminum and/or stainless steel.

In general, the platform **6212** is configured to be received by a heating device. More specifically, the platform **6212** is generally configured such that thermal transfer occurs between the heating elements of a heating device and the platform **6212**. The heat transferred to the platform **6212** is then further transferred to a substrate, as will be discussed in greater detail below.

The plurality of members **6318** can be dimensioned to be received by different regions of a heating device. For example, in some embodiments, the plurality of members **6318** can be dimensioned to be received within individual sample wells of a thermocycler (for example, thermocycler wells that are generally dimensioned to receive small volume test tubes such as 200 μL tubes). In general, the platform **6212** can include any number of members **6318**. For example, in some embodiments, the platform **6212** includes between 4 members and 96 members. Typically, heat is more evenly transferred to the platform **6212** when the number of members **6318** is larger, and they are distributed relatively evenly on the surface **6314**.

The support member **6220** can extend from the platform **6212** and be configured to couple to the substrate holder **6150**. In some embodiments, the support member **6220** can include one or more recesses or protrusions that couple to one or more complementary protrusions or recesses on the substrate holder **6150** to aid in coupling the substrate holder **6150** to the plate **6110**. In some embodiments, the support member **6220** can be entirely received by a portion of the substrate holder **6150**. In some embodiments, a portion of the support member **6220** can be received by a portion of the substrate holder **6150**. In some embodiments, the support member **6220** can be substantially flat on a top face. In some embodiments, the support member **6220** can be sized and positioned such that the plate **6110** can include multiple support members **6220**. For example, in some embodiments, the plate **6110** can include two support members **6220**.

Referring generally to FIGS. **16-22**, a substrate holder **6150** can include a bottom member **6460** and a top member **6480**. In some embodiments, the substrate holder **6150** can further include a slide **6452**. In some embodiments, the substrate holder **6150** can include a gasket **6454** (see also FIG. **21**).

Referring to FIGS. **17**A and **17**B, the bottom member **6460** can include a base **6562**, a first side wall **6566** and a second side wall **6568**. The bottom member **6460** can be configured to mount slide **6452**.

In some embodiments, the base **6562** can include a marker **6564**. The marker **6564** can aid in coupling the bottom member **6460** and the top member **6480** in a certain orientation. For example, in some embodiments, the bottom member **6460** and the top member **6480** may only be able to couple together in a single orientation. In some embodiments, the marker **6564** can be a line, groove, indent, protrusion, symbol, color, etc. to distinguish one side of the base **6562** from another side of the base **6562**.

The first side wall **6566** and the second side wall **6568** can be positioned on opposing longitudinal sides of the base **6562**. The side walls **6566** and **6568** can extend substantially perpendicular from the base **6562**. In some embodiments, the side walls **6566** and **6568** are slightly offset on edges of the base **6562** such that a portion of the base **6562** is exposed on either side of both side walls **6566** and **6568**. The side walls **6566** and **6568** can aid in securing the slide **6452** in the bottom member **6460**. In some embodiments, when the slide **6452** is secured in the bottom member **6460**, the slide **6452** may rest on the base **6562**. In some embodiments, the side walls **6566** and **6568** can be configured to engage with the top member **6480**. For example, in some embodiments, the side walls **6566** and **6568** can include one or more recesses **6570**. In some embodiments, the side wall **6566** can include a single recess **6570**, while the side wall **6568** includes multiple recesses (e.g., two recesses). Such a configuration can provide single direction coupling of the bottom member **6460** and the top member **6480** for ease of use.

The base **6562** can further include means for mounting the slide **6452**. For example, in some embodiments, the base **6562** can include a fastener housing **6572** and an end securing member **6574**. The end securing member **6574** can aid in securing the slide **6452** in the bottom member **6460**. In some embodiments, the end securing member **6574** can include a ridge to limit movement of the slide **6452** away from the base **6562**. The fastener housing **6572** can provide housing for a fastener **6458** (see e.g., FIG. **20**). The fastener **6458** can be a clip, ridge, or other means of removably coupling the slide **6452** to the bottom member **6460**. In some embodiments, the fastener **6458** can include a spring such that pushing the slide **6452** against the fastener **6458** causes the fastener **6458** to move past the ridge of the end securing member **6574**, allowing the slide **6452** to enter the bottom member **6460**. Once the slide **6452** is released, the spring can return the fastener **6458** to a more neutral position, causing the slide **6452** to abut the end securing member **6574**.

The bottom member **6460** can include means for coupling to the support member **6220** of the plate **6110**. For example, in some embodiments, the base **6562** can include an aperture **6576** extending through the base **6562** within the side walls **6566** and **6568**. Referring to FIGS. **25** and **26**, the bottom member **6460** is shown coupled to the plate **6110**. In some embodiments, the aperture **6576** can be sized such that the support member **6220** substantially fills the aperture **6576** (see FIG. **18**). In some embodiments, the aperture **6576** can be larger than the support member **6822**, such that the support member **6220** only fills a portion of aperture **6576** (see FIG. **19**). In some embodiments, the support member **6220**, the bottom member **6460** and the aperture **6576** are configured such that the slide **6452** is in close proximity to the support member **6220**. In some embodiments, the support member **6220**, the bottom member **6460** and the aperture **6576** are configured such that the slide **6452** is in direct contact with the support member **6220**. In some embodiments, the support member **6220**, the bottom member **6460** and the aperture **6576** are configured such that a portion of a sample region of the slide **6452** is in proximity to the support member **6220**. In some embodiments, the portion of the sample region of the slide **6452** is 50% to 100% of the sample region. In some embodiments, the portion of the sample region is at least 60% of the sample region. In some embodiments, the portion of the sample region is at least 75% of the sample region. In some embodiments, the portion of the sample region is at least 80% of the sample region. In some embodiments, the portion of the sample region is at least 85% of the sample region.

Referring back to FIGS. **16-22**, the bottom member **6460** can include an engagement mechanism for coupling the bottom member **6460** to the top member **6480**. In some embodiments, the engagement mechanism includes screws **6496**. Accordingly, the base **6562** can include one or more threaded apertures **6578** configured to receive the screws **6496**. The base **6562** can be sized such that the screws **6496** do not protrude underside of the base **6562**.

The gasket **6454** can be positioned inside the substrate holder **6150**. The gasket **6454** can include a plurality of apertures **7156** (see also FIG. **21**) that can create a plurality of wells when the gasket **6454** abuts the slide **6452**. In some embodiments, the gasket **6454** can be made of rubber, silicone, or a similar material to create a seal with the slide **6452**. In some embodiments, the gasket **6454** can be made of a material that is hydrophobic. Accordingly, different reactions can be conducted in the various wells of the gasket **6454**. In some embodiments, the engagement of the bottom member **6460** and the top member **6480** creates ample pressure to maintain division between the wells created by the apertures **7156**.

Referring to FIGS. **22**A and **22**B, the top member **6480** can include a body **7282** with walls **7284**. In some embodiments, one side of the body **7282** and/or the wall **7284** can include a marker **7286**. Marker **7286** can aid in coupling the bottom member **6460** and the top member **6480** in a certain orientation. For example, in some embodiments, the bottom member **6460** and the top member **6480** may only be able to couple together in a single orientation. In some embodiments, the marker **7286** can be a line, groove, indent, protrusion, symbol, color, etc. to distinguish one side of the body **7282** from another side of the body **7282**.

In some embodiments, the walls **7284** can be configured to engage with the bottom member **6460**. For example, in some embodiments, the walls **7284** can include one or more protrusions **7388**. In some embodiments, a first wall **7284** can include a single protrusion **7388**, while a second wall **7284** can includes multiple protrusions **7388** (e.g., two protrusions). Such a configuration can provide single direction coupling of the bottom member **6460** and the top member **6480** for ease of use.

The body **7282** can further include an engagement mechanism for coupling the top member **6480** to the bottom member **6460**. In some embodiments, the engagement mechanism includes screws **6496**. Accordingly, the body **7282** can include one or more threaded apertures **7290** configured to receive the screw **6496**. The body **7282** can be configured such that when the apertures **7290** receive the screws **6496**, the heads of the screws **6496** are flush or lower than an upper face of the body **7282**. Accordingly, the top member **6480** can be configured such that a top plate of a heating device can abut the top member **6480**, providing heating of the substrate holder **6150**.

In some embodiments, the body **7282** can include a recess **7392** on the underside of the body **7282**. In some embodiments, the recess **7392** can receive the gasket **6454**. In some embodiments, the recess **7392** can removably couple the gasket **6454**. In some embodiments, the body **7282** can include the gasket **6454**, such that the gasket **6454** is integrated with the body **7282**.

In some embodiments, the body **7282** can include a plurality of apertures **7294**. In some embodiments, the plurality of apertures **7294** can be configured to enable reagents to be added to the substrate on the slide **6452**. In some embodiments, the plurality of apertures **7294** can be configured to be aligned with the plurality of apertures **7156** of the gasket **6454**. In some embodiments, the plurality of apertures **7294** can be configured with a format and spacing to enable use with a multichannel pipette. In some embodiments, the plurality of apertures **7294** can be located on only a portion of the body **7282**. In some embodiments, the body **7282** can include labels or markings adjacent to the plurality of apertures **7294**.

While the substrate holder **6150** is described as including multiple pieces (e.g., the bottom member **6460**, the top member **6480**, the slide **6452**, the gasket **6454**, etc.), components of the substrate holder **6150** can be integrated with one another. For example, in some embodiments, the gasket **6454** can be integrated with the top member **6480**. As another example, the top member **6480** and the bottom member **6460** can be a single piece that is configured to receive the slide **6452**.

In some embodiments, the substrate holder **6150** can be made of a material that is reusable. For example, in some embodiments, the substrate holder **6150** can be washed and sanitized for reuse. Optionally, the gasket **6454** can be reusable or replaceable in such an embodiment. In some embodiments, the substrate holder **6150** can be made for single use and can be disposable.

Referring to FIGS. **23-25**, another example substrate cassette is described. An example device **7498** can be used to replace the substrate cassettes **4010**, **4810** described herein. The device **7498** can include a substrate holder **7400**, a gasket **6454**, and a substrate, such as a glass slide **6452**. The slide **6452** includes a first surface and a second surface. In some embodiments, the substrate holder **7400** is configured to receive the slide **6452**. In some embodiments, the substrate holder **7400** includes an attachment mechanism to hold the slide **6452** to the substrate holder **7400**. The second surface of the slide **6452** can provide a substrate for receiving a sample. In some embodiments, the substrate holder **7400** is plastic component (e.g., injection molded plastic component). In some embodiments, the gasket **6454** is configured to be positioned in between the substrate holder

7400 and the slide **6452**. In some embodiments, the device **7498** (or any one of its components) can be a single-use device (or component). In some embodiments, the device **7498** is entirely disposable or at least partially composed of disposable components.

FIG. **23**A shows the device **7498** in an assembled state. In particular, FIG. **23**A shows the substrate holder **7400** receiving the gasket **6454** and the slide **6452**. The substrate holder **7400** has longitudinal sides **7405** and latitudinal sides **7407**. The substrate holder **7400** can include one or more fasteners, such as a side mounted press latch **7410** for snap engagement. Any type of fastener that allows releasable engagement can be used, such as, for example, screws and press fit type connectors. The press latch **7410** can be mounted in a longitudinal side **7505** of the substrate holder **7400**, as shown in FIGS. **23**A-**23**C. Alternatively, in some embodiments, the press latch **7410** can be mounted in a latitudinal side **7407** of the substrate holder **7400**. In some embodiments, the substrate holder **7400** can include two, three, or four press latches. The press latch **7410** can be configured to engage the slide **6452**. In some embodiments, the press latch **7410** can be a lever, a clip, or a clamp. In some embodiments, the press latch **7410** can further include one or more springs.

The substrate holder **7400** can further include one or more engagement features, such as a first tab **7412***a* and a second tab **7412***b*. The first and second tabs **7412***a* and **7412***b*, respectively, can protrude from a longitudinal side **7505** of the substrate holder **7400** that opposes the longitudinal side having the press latch **7410**. In some embodiments, the substrate holder **7400** includes three, four, five, six, seven, eight, nine, ten or more tabs. In some embodiments, the tabs protrude from a longitudinal side **7505** or a latitudinal side **7407** of the substrate holder **7400**. The first and second tabs **7412***a* and **7412***b* can be configured to engage the slide **6452**. In some embodiments, the first and second tabs **7412***a* and **7412***b* are rigid and do not flex when engaging the slide **6452**. In some embodiments, the tabs **7412***a* and **7412***b* can be flexible.

Referring to FIG. **23**B, the substrate holder **7400** includes a bottom surface **7401**. The bottom surface **7401** includes a plurality of latitudinal ribs **7414** and longitudinal ribs **7416** configured to support the slide **6452** and the gasket **6454** when the device **7498** is assembled. The bottom surface **7401** further defines a plurality of apertures **7294** that are configured to align with the plurality of apertures **7156** defined by the gasket **6454**, when the device **7498** is assembled. The substrate holder **7400** further includes a c-shaped tab **7422** shaped to provide the user with an ergonomic grip surface to help facilitate engagement with the press latch **7410**.

Referring to FIG. **23**C, the slide **6452** includes a first surface **7446**, a second surface **7448**, and a side edge **7418**. When inserting the slide **6452** into the substrate holder **7400**, the side edge **7418** can be inserted first such that first and second tabs **7412***a* and **7412***b* engage the side edge **7418** as the slide **6452** rests on the gasket **6454** an on the plurality of latitudinal ribs **7414**. In some embodiments, the side **7418** can measure about 6 inches. In some embodiments, the sides shorter than the side **7418** can measure about 1 inch. In some embodiments, the slide **6452** can measure about 75 millimeters (mm) by 25 mm. In some embodiments, the slide **6452** can measure about 75 millimeters (mm) by 50 mm. In some embodiments, the slide **6452** can measure about 48 millimeters (mm) by 28 mm. In some embodiments, the slide **6452** can measure about 46 millimeters (mm) by 27 mm. In some embodiments, the slide **6452** is a glass slide.

Referring to FIG. **24**A, the substrate holder **7400** includes a top surface **7503**. The top surface **7503** defines the plurality of apertures **7294**. Furthermore, the top surface **7503** includes a logo **7502**, a first identifier **7504**, and a second identifier **7506**. The first identifier **7504** can identify the columns of the plurality of apertures **7294**. The second identifier **7506** can identify the rows of the plurality of apertures **7294**. In some embodiments, the first identifier **7504** and the second identifier **7506** are letters or numbers. In some embodiments, the first identifier **7504** and the second identifier **7506** can aid in aligning and/or inserting the slide **6452** into the substrate holder **7400** in a certain orientation. In some embodiments, the slide **6452** may include samples (e.g., biological material samples) on a portion of its surface that align with one or more of the plurality of apertures **7294**. In some embodiments, the samples (e.g., biological material samples) may be identified in the same manner as the corresponding aperture of the plurality of apertures **7294**. For example, in some embodiments, the slide **6452** may include a sample named "A1" that corresponds with the aperture **7511** labeled as "A1" by the first identifier **7504** and the second identifier **7506** in FIG. **24**A. As such, in some embodiments, the first identifier **7504** and the second identifier **7506** guide a user to correctly place the slide **6452** into the substrate holder **7400**. In some embodiments, the first identifier **7504** and the second identifier **7506** can be a line, groove, indent, protrusion, symbol, color, etc. to distinguish the rows and columns of the plurality of apertures **7294**.

Referring to FIG. **24**B, the substrate holder **7400** includes a first latitudinal rib **7414***a*, a second latitudinal rib **7414***b*, a third latitudinal rib **7414***c*, a fourth latitudinal rib **7414***d*, and a fifth latitudinal rib **7414***e* having a width w. The plurality of latitudinal ribs **7414** is configured to support a slide **6452**. The first, second, third, fourth, and fifth latitudinal ribs **7414***a*, **7414***b*, **7414***c*, **7414***d*, and **7414***e* can have an equal width w, as shown in FIG. **24**B. In some embodiments, the widths of the plurality of latitudinal ribs can vary. The first, second, and third latitudinal ribs **7414***a*, **7414***b*, and **7414***c*, respectively, extend perpendicular from the bottom surface **7401** near a first end **7515***a* of the substrate holder **7400**. The fifth latitudinal rib **7414***e* extends substantially perpendicular from the bottom surface **7401** near a second end **7515***b* of the substrate holder **7400**. In some embodiments, the substrate holder **7400** may include 6, 7, 8, 9, 10, 15, 20 or more latitudinal ribs.

The substrate holder **7400** further includes a first longitudinal rib **7416***a*, a second longitudinal rib **7416***b*, a third longitudinal rib **7416***c*, and a fourth longitudinal rib **7416***d* that extend substantially perpendicular from the bottom surface **7401**. The plurality of longitudinal ribs **7416** is configured to provide longitudinal support to the gasket **6454**. For example, in some embodiments, the plurality of longitudinal ribs **7416** abuts the longitudinal sides of the gasket **6454**. Furthermore, together with the fourth and fifth latitudinal ribs **7414***c* and **7414***e*, the plurality of longitudinal ribs frame an area of the bottom surface **7401** (e.g., gasket area) that is sufficiently sized and configured to receive the gasket **6454**. The first longitudinal rib **7416***a*, second longitudinal rib **7416***b*, third longitudinal rib **7416***c*, and fourth longitudinal rib **7416***d* can be disposed parallel to the longitudinal sides **7505** along the side edges. The first longitudinal rib **7416***a*, second longitudinal rib **7416***b*, third longitudinal rib **7416***c*, and fourth longitudinal rib **7416***d* have a second height **7426**, as shown in FIG. **24**B. The first, second, third, fourth, and fifth latitudinal ribs **7414***a*, **7414***b*, **7414***c*, **7414***d*, and **7414***e* have a first height **7424**, as shown in FIG. **32**B. The first height **7424** can be greater than the second height **7426**, as shown in FIGS. **32-33**. In some embodiments, the first height **7424** is equal to the second height **7426** (see FIG. **23**B). In some embodiments, the first height **7424** is less than the second height **7426**. The first and second longitudinal ribs **7416**_a_ and **7416**_b_ have a length l'. The third and fourth longitudinal ribs **7416**_c_ and **7416**_d_ have a length l''. The length l' is can be greater than the length l'', as shown in FIG. **23**B. In some embodiments, the length l' is equal to the length l''. In some embodiments, the length l' is less than the length l''. In some embodiments, the substrate holder **7400** may include 5, 6, 7, 8, 9, 10, 15, 20 or more longitudinal ribs.

The substrate holder **7400** further includes a third identifier **7508** that aids a user in positioning and/or orienting the substrate holder **7400**. For example, in some embodiments, the third identifier **7508** aids in identifying the first end **7515**_a_ or aids in distinguishing the first end **7515**_a_ from the second end **7515**_b_. Still yet in further embodiments, the third identifier can be a line, groove, indent, protrusion, symbol, color, etc. that aids a user in correctly positioning slide **6452** into the substrate holder **7400**.

Referring to FIG. **24**C, the substrate holder includes a first flexible tab **7513**_a_ and a second flexible tab **7513**_b_. The first and second flexible tabs **7513**_a_ and **7513**_b_, respectively, can protrude from top portions **7517** of the press latch **7410** that oppose the longitudinal side **7505** having the first and second tabs **7412**_a_ and **7412**_b_. The first and second flexible tabs **7513**_a_ and **7513**_b_ can protrude from an interior surface **75109** of the top portions **7517** of the press latch **7410**. To utilize the substrate holder **7400**, a user can grip the substrate holder **7400** in one hand with the bottom surface **7401** face up (i.e., facing the user). Next, the user can place the gasket **6454** over the plurality of apertures **7294**. The user can depress the press latch **7410** with one hand, this flexes the first and second flexible tabs **7513**_a_ and **7513**_b_ on one longitudinal side **7505** open, in the direction of arrows A. As such, the user can load the slide **6452** into the first and second tabs **7412**_a_ and **7412**_b_ first, and subsequently hinge slide **6452** into opposite longitudinal side (i.e., the side having the first and second flexible tabs **7513**_a_ and **7513**_b_). Lastly, the user can release the press latch **7410** so the slide **6452** can snap into the first and second flexible tabs **7513**_a_ and **7513**_b_.

Referring to FIG. **25**, the gasket **6454** includes a plurality of apertures **7156**. In some embodiments, the gasket **6454** includes eight apertures. In some embodiments, the gasket **6454** includes sixteen apertures. In some embodiments, the gasket **6454** includes 24 apertures. In some embodiments, the gasket **6454** includes 96 apertures. In some embodiments, the gasket **6454** is made from a material that can withstand temperatures up to about 60 degrees Celsius. In some embodiments, the gasket **6454** is made from a heat-resistant material. In some embodiments, the gasket **6454** is made from a flexible or pliable material. Non-limiting examples of flexible or pliable materials include rubber, silicone, and polyurethane. It is contemplated that the number of apertures is the same as the number of sample arrays on a substrate.

Spatial analysis methodologies and compositions described herein can provide a vast amount of analyte and/or expression data for a variety of analytes within a biological sample at high spatial resolution, while retaining native spatial context. Spatial analysis methods and compositions can include, e.g., the use of a capture probe including a spatial barcode (e.g., a nucleic acid sequence that provides information as to the location or position of an analyte within a cell or a tissue sample (e.g., mammalian cell or a mammalian tissue sample) and a capture domain that is capable of binding to an analyte (e.g., a protein and/or a nucleic acid) produced by and/or present in a cell. Spatial analysis methods and compositions can also include the use of a capture probe having a capture domain that captures an intermediate agent for indirect detection of an analyte. For example, the intermediate agent can include a nucleic acid sequence (e.g., a barcode) associated with the intermediate agent. Detection of the intermediate agent is therefore indicative of the analyte in the cell or tissue sample.

Non-limiting aspects of spatial analysis methodologies and compositions are described in U.S. Pat. Nos. 10,774,374, 10,724,078, 10,480,022, 10,059,990, 10,041,949, 10,002,316, 9,879,313, 9,783,841, 9,727,810, 9,593,365, 8,951,726, 8,604,182, 7,709,198, U.S. Patent Application Publication Nos. 2020/239946, 2020/080136, 2020/0277663, 2020/024641, 2019/330617, 2019/264268, 2020/256867, 2020/224244, 2019/194709, 2019/161796, 2019/085383, 2019/055594, 2018/216161, 2018/051322, 2018/0245142, 2017/241911, 2017/089811, 2017/067096, 2017/029875, 2017/0016053, 2016/108458, 2015/000854, 2013/171621, WO 2018/091676, WO 2020/176788, Rodriques et al., Science 363(6434):1463-1467, 2019; Lee et al., Nat. Protoc. 10(3):442-458, 2015; Trejo et al., PLoS ONE 14(2): e0212031, 2019; Chen et al., Science 348(6233):aaa6090, 2015; Gao et al., BMC Biol. 15:50, 2017; and Gupta et al., Nature Biotechnol. 36:1197-1202, 2018; the Visium Spatial Gene Expression Reagent Kits User Guide (e.g., Rev C, dated June 2020), and/or the Visium Spatial Tissue Optimization Reagent Kits User Guide (e.g., Rev C, dated July 2020), both of which are available at the 10× Genomics Support Documentation website, and can be used herein in any combination. Further non-limiting aspects of spatial analysis methodologies and compositions are described herein.

Some general terminology that may be used in this disclosure can be found in Section (I)(b) of WO 2020/176788 and/or U. S. Patent Application Publication No. 2020/0277663. Typically, a "barcode" is a label, or identifier, that conveys or is capable of conveying information (e.g., information about an analyte in a sample, a bead, and/or a capture probe). A barcode can be part of an analyte, or independent of an analyte. A barcode can be attached to an analyte. A particular barcode can be unique relative to other barcodes. For the purpose of this disclosure, an "analyte" can include any biological substance, structure, moiety, or component to be analyzed. The term "target" can similarly refer to an analyte of interest.

Analytes can be broadly classified into one of two groups: nucleic acid analytes, and non-nucleic acid analytes. Examples of non-nucleic acid analytes include, but are not limited to, lipids, carbohydrates, peptides, proteins, glycoproteins (N-linked or O-linked), lipoproteins, phosphoproteins, specific phosphorylated or acetylated variants of proteins, amidation variants of proteins, hydroxylation variants of proteins, methylation variants of proteins, ubiquitylation variants of proteins, sulfation variants of proteins, viral proteins (e.g., viral capsid, viral envelope, viral coat, viral accessory, viral glycoproteins, viral spike, etc.), extracellular and intracellular proteins, antibodies, and antigen binding fragments. In some embodiments, the analyte(s) can be localized to subcellular location(s), including, for example, organelles, e.g., mitochondria, Golgi apparatus, endoplasmic reticulum, chloroplasts, endocytic vesicles, exocytic vesicles, vacuoles, lysosomes, etc. In some embodiments, analyte(s) can be peptides or proteins, including without limitation antibodies and enzymes. Additional examples of analytes can be found in Section (I)(c) of WO 2020/176788 and/or U.S. Patent Application Publication No. 2020/0277663. In some embodiments, an analyte can be detected indirectly, such as through detection of an intermediate agent, for example, a ligation product or an analyte capture agent (e.g., an oligonucleotide-conjugated antibody), such as those described herein.

A "biological sample" is typically obtained from the subject for analysis using any of a variety of techniques including, but not limited to, biopsy, surgery, and laser capture microscopy (LCM), and generally includes cells and/or other biological material from the subject. In some embodiments, a biological sample can be a tissue section. In some embodiments, a biological sample can be a fixed and/or stained biological sample (e.g., a fixed and/or stained tissue section). Non-limiting examples of stains include histological stains (e.g., hematoxylin and/or eosin) and immunological stains (e.g., fluorescent stains). In some embodiments, a biological sample (e.g., a fixed and/or stained biological sample) can be imaged. Biological samples are also described in Section (I)(d) of WO 2020/176788 and/or U.S. Patent Application Publication No. 2020/0277663.

In some embodiments, a biological sample is permeabilized with one or more permeabilization reagents. For example, permeabilization of a biological sample can facilitate analyte capture. Exemplary permeabilization agents and conditions are described in Section (I)(d)(ii)(13) or the Exemplary Embodiments Section of WO 2020/176788 and/or U.S. Patent Application Publication No. 2020/0277663.

Array-based spatial analysis methods involve the transfer of one or more analytes from a biological sample to an array of features on a substrate, where each feature is associated with a unique spatial location on the array. Subsequent analysis of the transferred analytes includes determining the identity of the analytes and the spatial location of the analytes within the biological sample. The spatial location of an analyte within the biological sample is determined based on the feature to which the analyte is bound (e.g., directly or indirectly) on the array, and the feature's relative spatial location within the array.

A "capture probe" refers to any molecule capable of capturing (directly or indirectly) and/or labelling an analyte (e.g., an analyte of interest) in a biological sample. In some embodiments, the capture probe is a nucleic acid or a polypeptide. In some embodiments, the capture probe includes a barcode (e.g., a spatial barcode and/or a unique molecular identifier (UMI)) and a capture domain). In some embodiments, a capture probe can include a cleavage domain and/or a functional domain (e.g., a primer-binding site, such as for next-generation sequencing (NGS)). See, e.g., Section (II)(b) (e.g., subsections (i)-(vi)) of WO 2020/176788 and/or U.S. Patent Application Publication No. 2020/0277663. Generation of capture probes can be achieved by any appropriate method, including those described in Section (II)(d)(ii) of WO 2020/176788 and/or U.S. Patent Application Publication No. 2020/0277663.

In some embodiments, more than one analyte type (e.g., nucleic acids and proteins) from a biological sample can be detected (e.g., simultaneously or sequentially) using any appropriate multiplexing technique, such as those described in Section (IV) of WO 2020/176788 and/or U.S. Patent Application Publication No. 2020/0277663.

In some embodiments, detection of one or more analytes (e.g., protein analytes) can be performed using one or more analyte capture agents. As used herein, an "analyte capture agent" refers to an agent that interacts with an analyte (e.g., an analyte in a biological sample) and with a capture probe (e.g., a capture probe attached to a substrate or a feature) to identify the analyte. In some embodiments, the analyte capture agent includes: (i) an analyte binding moiety (e.g., that binds to an analyte), for example, an antibody or antigen-binding fragment thereof, (ii) analyte binding moiety barcode; and (iii) an analyte capture sequence. As used herein, the term "analyte binding moiety barcode" refers to a barcode that is associated with or otherwise identifies the analyte binding moiety. As used herein, the term "analyte capture sequence" refers to a region or moiety configured to hybridize to, bind to, couple to, or otherwise interact with a capture domain of a capture probe. In some cases, an analyte binding moiety barcode (or portion thereof) may be able to be removed (e.g., cleaved) from the analyte capture agent. Additional description of analyte capture agents can be found in Section (II)(b)(ix) of WO 2020/176788 and/or Section (II)(b)(viii) U.S. Patent Application Publication No. 2020/0277663.

There are at least two methods to associate a spatial barcode with one or more neighboring cells, such that the spatial barcode identifies the one or more cells, and/or contents of the one or more cells, as associated with a particular spatial location. One method is to promote analytes or analyte proxies (e.g., intermediate agents) out of a cell and towards a spatially-barcoded array (e.g., including spatially-barcoded capture probes). Another method is to cleave spatially-barcoded capture probes from an array and promote the spatially-barcoded capture probes towards and/or into or onto the biological sample.

In some cases, capture probes may be configured to prime, replicate, and consequently yield optionally barcoded extension products from a template (e.g., a DNA or RNA template, such as an analyte or an intermediate agent (e.g., a ligation product or an analyte capture agent), or a portion thereof), or derivatives thereof (see, e.g., Section (II)(b)(vii) of WO 2020/176788 and/or U.S. Patent Application Publication No. 2020/0277663 regarding extended capture probes). In some cases, capture probes may be configured to form ligation products with a template (e.g., a DNA or RNA template, such as an analyte or an intermediate agent, or portion thereof), thereby creating ligations products that serve as proxies for a template.

Additional variants of spatial analysis methods, including in some embodiments, an imaging step, are described in Section (II)(a) of WO 2020/176788 and/or U.S. Patent Application Publication No. 2020/0277663. Analysis of captured analytes (and/or intermediate agents or portions thereof), for example, including sample removal, extension of capture probes, sequencing (e.g., of a cleaved extended capture probe and/or a cDNA molecule complementary to an extended capture probe), sequencing on the array (e.g., using, for example, in situ hybridization or in situ ligation approaches), temporal analysis, and/or proximity capture, is described in Section (II)(g) of WO 2020/176788 and/or U.S. Patent Application Publication No. 2020/0277663. Some quality control measures are described in Section (II)(h) of WO 2020/176788 and/or U.S. Patent Application Publication No. 2020/0277663.

Spatial information can provide information of biological and/or medical importance. For example, the methods and compositions described herein can allow for: identification of one or more biomarkers (e.g., diagnostic, prognostic, and/or for determination of efficacy of a treatment) of a disease or disorder; identification of a candidate drug target for treatment of a disease or disorder; identification (e.g., diagnosis) of a subject as having a disease or disorder; identification of stage and/or prognosis of a disease or disorder in a subject; identification of a subject as having an increased likelihood of developing a disease or disorder; monitoring of progression of a disease or disorder in a subject; determination of efficacy of a treatment of a disease or disorder in a subject; identification of a patient subpopulation for which a treatment is effective for a disease or disorder; modification of a treatment of a subject with a disease or disorder; selection of a subject for participation in a clinical trial; and/or selection of a treatment for a subject with a disease or disorder.

Spatial information can provide information of biological importance. For example, the methods and compositions described herein can allow for: identification of transcriptome and/or proteome expression profiles (e.g., in healthy and/or diseased tissue); identification of multiple analyte types in close proximity (e.g., nearest neighbor analysis); determination of up- and/or down-regulated genes and/or proteins in diseased tissue; characterization of tumor microenvironments; characterization of tumor immune responses; characterization of cells types and their co-localization in tissue; and identification of genetic variants within tissues (e.g., based on gene and/or protein expression profiles associated with specific disease or disorder biomarkers).

Typically, for spatial array-based methods, a substrate functions as a support for direct or indirect attachment of capture probes to features of the array. A "feature" is an entity that acts as a support or repository for various molecular entities used in spatial analysis. In some embodiments, some or all of the features in an array are functionalized for analyte capture. Exemplary substrates are described in Section (II)(c) of WO 2020/176788 and/or U.S. Patent Application Publication No. 2020/0277663. Exemplary features and geometric attributes of an array can be found in Sections (II)(d)(i), (II)(d)(iii), and (II)(d)(iv) of WO 2020/176788 and/or U.S. Patent Application Publication No. 2020/0277663.

Generally, analytes and/or intermediate agents (or portions thereof) can be captured when contacting a biological sample with a substrate including capture probes (e.g., a substrate with capture probes embedded, spotted, printed, fabricated on the substrate, or a substrate with features (e.g., beads, wells) comprising capture probes). As used herein, "contact," "contacted," and/or "contacting," a biological sample with a substrate refers to any contact (e.g., direct or indirect) such that capture probes can interact (e.g., bind covalently or non-covalently (e.g., hybridize)) with analytes from the biological sample. Capture can be achieved actively (e.g., using electrophoresis) or passively (e.g., using diffusion). Analyte capture is further described in Section (II)(e) of WO 2020/176788 and/or U.S. Patent Application Publication No. 2020/0277663.

In some cases, spatial analysis can be performed by attaching and/or introducing a molecule (e.g., a peptide, a lipid, or a nucleic acid molecule) having a barcode (e.g., a spatial barcode) to a biological sample (e.g., to a cell in a biological sample). In some embodiments, a plurality of molecules (e.g., a plurality of nucleic acid molecules) having a plurality of barcodes (e.g., a plurality of spatial barcodes) are introduced to a biological sample (e.g., to a plurality of cells in a biological sample) for use in spatial analysis. In some embodiments, after attaching and/or introducing a molecule having a barcode to a biological sample, the biological sample can be physically separated (e.g., dissociated) into single cells or cell groups for analysis. Some such methods of spatial analysis are described in Section (III) of WO 2020/176788 and/or U.S. Patent Application Publication No. 2020/0277663.

In some cases, spatial analysis can be performed by detecting multiple oligonucleotides that hybridize to an analyte. In some instances, for example, spatial analysis can be performed using RNA-templated ligation (RTL). Methods of RTL have been described previously. See, e.g., Credle et al., *Nucleic Acids Res.* 2017 Aug. 21; 45(14):e128. Typically, RTL includes hybridization of two oligonucleotides to adjacent sequences on an analyte (e.g., an RNA molecule, such as an mRNA molecule). In some instances, the oligonucleotides are DNA molecules. In some instances, one of the oligonucleotides includes at least two ribonucleic acid bases at the 3' end and/or the other oligonucleotide includes a phosphorylated nucleotide at the 5' end. In some instances, one of the two oligonucleotides includes a capture domain (e.g., a poly(A) sequence, a non-homopolymeric sequence). After hybridization to the analyte, a ligase (e.g., Splint® ligase) ligates the two oligonucleotides together, creating a ligation product. In some instances, the two oligonucleotides hybridize to sequences that are not adjacent to one another. For example, hybridization of the two oligonucleotides creates a gap between the hybridized oligonucleotides. In some instances, a polymerase (e.g., a DNA polymerase) can extend one of the oligonucleotides prior to ligation. After ligation, the ligation product is released from the analyte. In some instances, the ligation product is released using an endonuclease (e.g., RNAse H). The released ligation product can then be captured by capture probes (e.g., instead of direct capture of an analyte) on an array, optionally amplified, and sequenced, thus determining the location and optionally the abundance of the analyte in the biological sample.

During analysis of spatial information, sequence information for a spatial barcode associated with an analyte is obtained, and the sequence information can be used to provide information about the spatial distribution of the analyte in the biological sample. Various methods can be used to obtain the spatial information. In some embodiments, specific capture probes and the analytes they capture are associated with specific locations in an array of features on a substrate. For example, specific spatial barcodes can be associated with specific array locations prior to array fabrication, and the sequences of the spatial barcodes can be stored (e.g., in a database) along with specific array location information, so that each spatial barcode uniquely maps to a particular array location.

Alternatively, specific spatial barcodes can be deposited at predetermined locations in an array of features during fabrication such that at each location, only one type of spatial barcode is present so that spatial barcodes are uniquely associated with a single feature of the array. Where necessary, the arrays can be decoded using any of the methods described herein so that spatial barcodes are uniquely associated with array feature locations, and this mapping can be stored as described above.

When sequence information is obtained for capture probes and/or analytes during analysis of spatial information, the locations of the capture probes and/or analytes can be determined by referring to the stored information that uniquely associates each spatial barcode with an array feature location. In this manner, specific capture probes and captured analytes are associated with specific locations in the array of features. Each array feature location represents a position relative to a coordinate reference point (e.g., an array location, a fiducial marker) for the array. Accordingly, each feature location has an "address" or location in the coordinate space of the array.

Some exemplary spatial analysis workflows are described in the Exemplary Embodiments section of WO 2020/176788 and/or U.S. Patent Application Publication No. 2020/0277663. See, for example, the Exemplary embodiment starting with "In some non-limiting examples of the workflows described herein, the sample can be immersed . . . " of WO 2020/176788 and/or U.S. Patent Application Publication No. 2020/0277663. See also, e.g., the Visium Spatial Gene Expression Reagent Kits User Guide (e.g., Rev C, dated June 2020), and/or the Visium Spatial Tissue Optimization Reagent Kits User Guide (e.g., Rev C, dated July 2020).

In some embodiments, spatial analysis can be performed using dedicated hardware and/or software, such as any of the systems described in Sections (II)(e)(ii) and/or (V) of WO 2020/176788 and/or U.S. Patent Application Publication No. 2020/0277663, or any of one or more of the devices or methods described in Sections Control Slide for Imaging, Methods of Using Control Slides and Substrates for, Systems of Using Control Slides and Substrates for Imaging, and/or Sample and Array Alignment Devices and Methods, Informational labels of WO 2020/123320.

Suitable systems for performing spatial analysis can include components such as a chamber (e.g., a flow cell or sealable, fluid-tight chamber) for containing a biological sample. The biological sample can be mounted for example, in a biological sample holder. One or more fluid chambers can be connected to the chamber and/or the sample holder via fluid conduits, and fluids can be delivered into the chamber and/or sample holder via fluidic pumps, vacuum sources, or other devices coupled to the fluid conduits that create a pressure gradient to drive fluid flow. One or more valves can also be connected to fluid conduits to regulate the flow of reagents from reservoirs to the chamber and/or sample holder.

The systems can optionally include a control unit that includes one or more electronic processors, an input interface, an output interface (such as a display), and a storage unit (e.g., a solid state storage medium such as, but not limited to, a magnetic, optical, or other solid state, persistent, writeable and/or re-writeable storage medium). The control unit can optionally be connected to one or more remote devices via a network. The control unit (and components thereof) can generally perform any of the steps and functions described herein. Where the system is connected to a remote device, the remote device (or devices) can perform any of the steps or features described herein. The systems can optionally include one or more detectors (e.g., CCD, CMOS) used to capture images. The systems can also optionally include one or more light sources (e.g., LED-based, diode-based, lasers) for illuminating a sample, a substrate with features, analytes from a biological sample captured on a substrate, and various control and calibration media.

The systems can optionally include software instructions encoded and/or implemented in one or more of tangible storage media and hardware components such as application specific integrated circuits. The software instructions, when executed by a control unit (and in particular, an electronic processor) or an integrated circuit, can cause the control unit, integrated circuit, or other component executing the software instructions to perform any of the method steps or functions described herein.

In some cases, the systems described herein can detect (e.g., register an image) the biological sample on the array. Exemplary methods to detect the biological sample on an array are described in PCT Application No. 2020/061064 and/or U.S. patent application Ser. No. 16/951,854.

Prior to transferring analytes from the biological sample to the array of features on the substrate, the biological sample can be aligned with the array. Alignment of a biological sample and an array of features including capture probes can facilitate spatial analysis, which can be used to detect differences in analyte presence and/or level within different positions in the biological sample, for example, to generate a three-dimensional map of the analyte presence and/or level. Exemplary methods to generate a two- and/or three-dimensional map of the analyte presence and/or level are described in PCT Application No. 2020/053655 and spatial analysis methods are generally described in WO 2020/061108 and/or U.S. patent application Ser. No. 16/951,864.

In some cases, a map of analyte presence and/or level can be aligned to an image of a biological sample using one or more fiducial markers, e.g., objects placed in the field of view of an imaging system which appear in the image produced, as described in the Substrate Attributes Section, Control Slide for Imaging Section of WO 2020/123320, PCT Application No. 2020/061066, and/or U.S. patent application Ser. No. 16/951,843. Fiducial markers can be used as a point of reference or measurement scale for alignment (e.g., to align a sample and an array, to align two substrates, to determine a location of a sample or array on a substrate relative to a fiducial marker) and/or for quantitative measurements of sizes and/or distances.

While this specification contains many specific implementation details, these should not be construed as limitations on the scope of any inventions or of what may be claimed, but rather as descriptions of features specific to particular implementations of particular inventions. Certain features that are described in this specification in the context of separate implementations can also be implemented in combination in a single implementation. Conversely, various features that are described in the context of a single implementation can also be implemented in multiple implementations separately or in any suitable sub-combination. Moreover, although features may be described above as acting in certain combinations and even initially claimed as such, one or more features from a claimed combination can in some cases be excised from the combination, and the claimed combination may be directed to a sub-combination or variation of a sub-combination.

Similarly, while operations are depicted in the drawings in a particular order, this should not be understood as requiring that such operations be performed in the particular order shown or in sequential order, or that all illustrated operations be performed, to achieve desirable results. In certain circumstances, multitasking and parallel processing may be advantageous. Moreover, the separation of various system components in the implementations described above should not be understood as requiring such separation in all implementations, and it should be understood that the described program components and systems can generally be integrated together in a single software product or packaged into multiple software products.

Thus, particular implementations of the subject matter have been described. Other implementations are within the scope of the following claims. In some cases, the actions recited in the claims can be performed in a different order and still achieve desirable results. In addition, the processes depicted in the accompanying figures do not necessarily require the particular order shown, or sequential order, to achieve desirable results. In certain implementations, multitasking and parallel processing may be advantageous.

What is claimed is:

1. A method for capturing analytes from one or more biological samples, the method comprising:

placing the one or more biological samples in contact with capture probes on a substrate, the one or more biological samples including the analytes;

inserting the substrate into a substrate holder of a substrate cassette;

subsequent to placing the one or more biological samples in contact with the capture probes on the substrate and inserting the substrate into the substrate holder, arranging the substrate cassette and the substrate to align one or more apertures of the substrate cassette with one or more substrate regions of the substrate and define one or more buffer chambers on the one or more substrate regions;

supplying buffers in the one or more buffer chambers;

arranging a cathode to place one or more electrode plates of the cathode within the one or more buffer chambers; and

generating electric fields between the one or more substrate regions and the one or more electrode plates to cause the analytes in the one or more biological samples to migrate toward the capture probes on the substrate, wherein the substrate functions as an anode that is removably coupled to the substrate cassette.

2. The method of claim **1**, wherein the capture probes are immobilized on the one or more substrate regions, and

wherein the one or more biological samples are placed in contact with the capture probes on the one or more substrate regions.

3. The method of claim **1**, wherein the one or more substrate regions include one or more wells recessed on the substrate.

4. The method of claim **1**,

wherein the substrate holder comprises a substrate mount for securing the substrate,

wherein the substrate cassette comprises:

a gasket including one or more gasket apertures configured to align with the one or more substrate regions when the substrate is secured by the substrate holder, wherein the one or more apertures include the one or more gasket apertures, and wherein the substrate holder comprises:

one or more holder apertures configured to align with the one or more gasket apertures when the substrate is secured by the substrate holder, wherein the one or more apertures include the one or more gasket apertures and the one or more holder apertures.

5. The method of claim **1**, further comprising:

providing a substrate cover including one or more cover apertures and mounting the cathode; and

placing the substrate cover onto the substrate cassette such that the one or more cover apertures of the substrate cover are aligned with the one or more apertures of the substrate cassette, respectively, and such that the one or more electrode plates of the cathode extends through the one or more cover apertures into the one or more apertures of the substrate.

6. The method of claim **1**, wherein the substrate is coated with a conductive material, and

wherein the conductive material includes at least one of tin oxide (TO), indium tin oxide (ITO), a transparent conductive oxide (TCO), aluminum doped zinc oxide (AZO), or fluorine doped tin oxide (FTO).

7. The method of claim **1**, further comprising:

arranging one or more spacers between the substrate and the cathode to define the one or more buffer chambers.

8. The method of claim **7**, wherein the one or more spacers comprise a non-conductive material.

9. The method of claim **1**, wherein said arranging the cathode to place the one or more electrode plates of the cathode within the one or more buffer chambers further comprises at least partially immersing the one or more electrode plates of the cathode in the buffers supplied in the one or more buffer chambers.

10. The method of claim **9**, wherein the buffers supplied in the one or more buffer chambers include a permeabilization reagent.

11. The method of claim **1**, wherein the buffers supplied in the one or more buffer chambers include at least one of a permeabilization reagent, a permeabilization buffer, a permeabilization enzyme, a buffer without a permeabilization reagent, a permeabilization gel, or a permeabilization solution.

12. The method of claim **1**, further comprising:

providing, from a power supply, a voltage between the substrate and the cathode through electrical wires connecting the power supply to the substrate and the cathode.

13. The method of claim **12**, further comprising:

connecting the electrical wires from the power supply to the substrate and the cathode prior to providing the voltage.

14. The method of claim **1**, wherein the one or more biological samples each comprises a cell or a tissue including a cell, and wherein the analytes comprise one of a protein or a nucleic acid.

15. The method of claim **1**, wherein the substrate cassette comprises a non-conductive material, and wherein the non-conductive material is one of plastic, glass, porcelain, or rubber.

16. The method of claim **1**, further comprising:

immersing at least a portion of the one or more electrode plates of the cathode in a buffer supplied in the one or more buffer chambers; and

providing a voltage between the substrate cassette and the one or more electrode plates to generate the electric fields between the one or more substrate regions and the one or more electrode plates.

17. The method of claim **16**, wherein the capture probes are immobilized on the one or more substrate regions, and

wherein the one or more biological samples are placed in contact with the capture probes on the one or more substrate regions.

18. The method of claim **17**, wherein the one or more substrate regions include one or more wells recessed on the substrate cassette.

19. The method of claim **16**, wherein the substrate cassette is coated with a conductive material, and

wherein the conductive material includes at least one of tin oxide (TO), indium tin oxide (ITO), a transparent conductive oxide (TCO), aluminum doped zinc oxide (AZO), or fluorine doped tin oxide (FTO).

20. The method of claim **16**, wherein the substrate cassette comprises a non-conductive material, and wherein the non-conductive material is one of plastic, glass, porcelain, or rubber.

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